Reporting AI-Assisted Proofs in Mathematical Risk Management: A Proposed Standard, with an Axiomatic Characterization of the Loss Ratio as a Case Study

Hirbod Assa*

Abstract

Mathematical results in risk management and actuarial science are increasingly obtained with substantial help from AI, from a chatbot that suggests a lemma to panels of AI agents that propose, prove and attack theorems. The declarations of AI use that journals now require do not tell a reader which parts of such a result were exposed to AI error and which checks they passed. This paper proposes a standard for reporting AI-assisted proofs: nine items, applied in proportion to how heavily AI was used, each marked by whether an editor or reviewer can verify it, and designed so that honest reporting is in the author's interest. Formal verification in a proof assistant and a public repository of the proof records are encouraged rather than required.

The standard is applied to a case study, carried out with a panel of AI agents under the author's direction: an axiomatic characterization of the loss ratio as a risk indicator, in the spirit of Cherny and Madan's theory of performance measures. The premium and the risk measure are implied by the loss ratio itself, the premium as the point at which the ratio stops being finite along the cash line $c \mapsto \mathrm{LR}(X+c)$. Axioms that refer only to values of the loss ratio – a non-finite point on each cash line, comonotone pooling of books with zero premium, law invariance, and continuity (respectively monotonicity) of the sure amounts that remove the premium and the loss of a book – characterize the loss ratios I_h/I_g, ratios of signed Choquet integrals (respectively distortion premia), with (h,g) unique; none of the axioms can be dropped, and no axiom on finite values alone can replace the pooling axiom. The report of the case study records the organization of the agents, the seventeen errors in AI-generated mathematics that were found, the checks applied, a formal check in Lean, the revisions prompted by readers, and what has not been verified. We compare the approach with recent large-scale multi-agent efforts in mathematics.

Keywords: AI-assisted proofs; reporting standards; large language models; multi-agent systems; formal verification; Lean; loss ratio; implied premium; axiomatic characterization; signed Choquet integral; comonotonic additivity.

1 Introduction

AI is now used in mathematical research in many ways. At one end, a researcher asks a chatbot for help with a lemma and records this in a line of the acknowledgements. At the other end, organizations run large numbers of AI agents that propose results, write proofs, attack them and, in some cases, check them in a proof assistant (Section 2.1). Between the two lies a mode of work that is within reach of an

*Department of Mathematics and Statistics, University College Dublin, and Model Library Ltd. *Draft of 9 November 2026.* The use of generative AI in this work is described in Section 8 and in the declaration before the references. The proofs have undergone AI-assisted critical review, which is fallible and does not constitute formal verification; the loss-ratio-specific steps of an earlier formulation have in addition been checked in Lean in an abstract setting. See Section 8.8.

individual researcher: a panel of AI agents, each with an assigned role, working on one problem under the direction of a mathematician who poses the questions, rejects formulations and decides on content. This paper is about how results obtained in this way should be reported.

The question is practical. A proof drafted by AI can read well and be wrong, and the errors found in the work reported here were mostly semantic: a correct argument for a slightly different statement, typically at a degenerate case, which careful reading can miss (Table 5). A reader therefore needs to know not only that AI was used, but for what, under whose direction, with which checks, which errors were found, and what has and has not been verified. The declarations of AI use that journals ask for record much less. At the same time, editors face a growing volume of submissions and a limited supply of reviewers, and a reporting requirement that adds to the reviewer's burden, or that cannot be checked, will not be adopted. A standard must therefore be proportionate, verifiable where possible, and useful to the reviewer.

The paper makes two contributions. The first is a proposed standard for reporting AI-assisted proofs in mathematical risk management and actuarial science (Section 2). It consists of nine items, from the tools used to the public records of the process. The items are applied in three tiers according to how heavily AI was used, so that a paper in which AI helped with editing is not asked for what a paper whose proofs were developed by agents must report. Each item is marked by whether an editor or reviewer can verify it, and the standard is designed so that reporting errors and checks honestly strengthens a paper rather than weakens it. Formal verification in a proof assistant and a public repository are encouraged, and their scope is to be reported precisely, but they are not required. To the author's knowledge, this is the first proposal of such a standard in our field. It is offered for the community to test and improve.

The second contribution is a case study to which the standard is applied in full: a mathematical result obtained with a panel of AI agents, together with the report of how it was obtained. The problem was chosen by the author as a test. It is not a textbook exercise, its answer was not known in advance, and its solution needs machinery that is not routine. Like most mathematicians, the author had long carried a few questions that seemed worth answering. One of them is to characterize the loss ratio as a risk indicator by necessary and sufficient conditions, in the way that Cherny and Madan (2009) characterized performance measures such as the Sharpe ratio and the RAROC through their theory of acceptability indices.

The loss ratio, claims divided by premium, is among the most widely used quantities in insurance. Ex post it is incurred claims over earned premium. Ex ante it is reported through its expected value, which can be misleading when the risk of a book is concentrated in the tail: two books with the same expected loss ratio may differ greatly in their exposure to large losses. A natural remedy is to report a tail measure of the claims, such as the value at risk or the expected shortfall, divided by the premium (Artzner et al., 1999). The loss ratio then becomes a model-based, or mark-to-model, quantity: a risk measure of the loss divided by a premium principle, of which the ex-post ratio is the special case in which the model is the realized outcome. The question is which properties are necessary and sufficient for a functional of the loss to be a loss ratio of this kind.

◇ **Discussion** (motivation of the problem). *When an earlier version was circulated, a reader asked publicly why one should characterize an accounting ratio axiomatically at all, and whether the choice of such a problem undermined the demonstration of what AI can do. In answering, the author and the main assistant separated the ex-post accounting ratio from the ex-ante ratio used in risk reporting, where a tail measure over the premium is a model-based quantity whose defining properties are not obvious. The paragraph above, and the statement that the problem was chosen deliberately as a test, are the outcome of that exchange.*

Our answer recovers the premium and the risk measure from the loss ratio itself. Along the *cash line* $c \mapsto \mathrm{LR}(X+c)$ of a book X there is exactly one point at which the ratio is not finite; the premium is minus this point, the sure payment at which the book changes hands, and the risk measure follows from the accounting identity loss = loss ratio × premium. With this reading, axioms that refer only to values of the loss ratio characterize the loss ratios that are ratios I_h/I_g of signed Choquet integrals

(Theorems 6.1 and 6.2). The central axiom states that the underwriting results of comonotone books of zero premium add when the books are pooled. None of the axioms can be dropped, and no axiom that depends only on finite values of the loss ratio can replace the pooling axiom (Section 7 and Proposition 5.3).

The case study is reported in Section 8, item by item against the proposed standard. Three facts from it shaped this paper. First, the central idea of the result, the reading of the premium and the risk measure from the loss ratio, arose from the author's objections to AI-drafted formulations that assumed what should have been proved. Second, Peng Liu, who read an earlier version, remarked that some of its axioms were stated through the implied premium and risk measure and asked for axioms stated more explicitly on the loss ratio itself; the present axioms were obtained in response, with a second panel of agents (Section 8.7). Third, an editor of an actuarial journal, asked for an opinion on the reporting standard, suggested that the actuarial part be condensed and that the standard be presented as a contribution in its own right, with the mathematical result as its case study; the paper has been restructured accordingly, and the proofs and secondary results have moved to a Supplement.

Several passages of the paper record exchanges between the author and the main AI assistant in the course of the work. They are set in italics and introduced by the mark $\diamond$ *Discussion*. They are formal summaries of what was discussed and concluded, not transcripts, and they are included so that the reader can see where the author's questions and objections changed the mathematics or the presentation.

The paper is organized as follows. Section 2 reviews current practice, compares the approach of this paper with large-scale multi-agent efforts, and proposes the reporting standard. Sections 3 to 7 present the mathematical case study: the related literature, the setting, the axioms, the main results and the independence of the axioms. Section 8 reports the AI process of the case study against the standard, including the formal check in Lean and the revisions. Section 9 discusses open problems on both sides. The Supplement contains the proofs, the version without law invariance, the version for nonnegative losses, further consequences, and the timed record of the Lean check.

2 AI-assisted proofs and a proposed reporting standard

This section describes current practice in the use of AI for mathematical proofs, compares the approach of this paper with recent large-scale efforts, and proposes the reporting standard. The standard is applied to the case study in Section 8.

2.1 Current practice

In most papers in our field that acknowledge AI, a general-purpose chatbot was asked for help with a step, a lemma, a computation or the wording, and the acknowledgement says so in a sentence. The proofs remain the author's, and a one-line declaration is adequate. A different mode of work has appeared in which AI agents, instances of a language model given a role, tools and a task, are organized to propose, prove, criticize and formalize results, and the human contribution is to choose problems, direct the work and decide what to accept.

The most visible examples are the releases of OpenAI in 2026, which illustrate several distinct ways of combining AI discovery with verification. In February, OpenAI submitted proof attempts by an internal model to First Proof, a set of ten research-level problems posed by experts in their fields (OpenAI, 2026b). The model was run "with limited human supervision": the researchers sometimes suggested retrying strategies that had appeared fruitful, asked the model to expand or clarify parts of a proof after expert feedback, facilitated a back-and-forth between the model and ChatGPT "for verification, formatting, and style", and for some problems presented the best of a few attempts selected by human judgment. One attempt first believed likely correct was later judged incorrect in the light of the official commentary, OpenAI states that the process was not as clean as a properly controlled evaluation,

and the accompanying preprint includes an appendix of prompt patterns. In August, OpenAI published ten results in mathematics and theoretical computer science, with certificates in the proof assistant Lean (OpenAI, 2026e). In September it announced a resolution of the Navier–Stokes problem, with a written proof and a Lean formalization (OpenAI, 2026d). OpenAI describes a system of coordinating agents, powered by an internal model, with access to a cached copy of the internet and to code execution, and subdivided into groups that communicate internally; the group that produced the result involved of the order of ten thousand concurrent agents. Different groups were prompted with different variants of the problem statement, two leading to a proof of regularity and two to a counterexample. A run on easier problems produced, with nearly a hundred agents over about fifty hours, a result on the Euler equations, after which the researchers moved agents to Navier–Stokes, prompted them with the Euler result, and switched them to a further trained version of the model. Groups were encouraged to explore diverse approaches and were then "cross-pollinated" by using a coding agent to consolidate the most useful insights of each group into follow-up prompts. The resolution came after about 88 hours and 2.7 million messages between agents, and the Lean formalization took another 17 hours. In October OpenAI released about seven hundred manuscripts produced by an internal model by one fixed procedure on some four thousand posed problems, together with a Lean library covering a part of the results and a change log recording retractions and repairs (OpenAI, 2026c); this release did not use the large communicating configuration of the Navier–Stokes run. The repository describes the manuscripts as "raw research data, rather than polished academic papers", and states that unformalized results could contain errors. Later in October, OpenAI announced three further results, in harmonic analysis, complexity theory and quantum field theory, again described its manuscripts as raw research data, and tested three ways of turning them into understanding: releasing the materials and leaving researchers to organize themselves; forming an external group of collaborators while releasing the materials; and forming an external group and leaving the release to it (OpenAI, 2026a). The collaborations follow four stated principles: transparency, no academic credit claimed by OpenAI for discoveries of its models, recognition of the groups for the understanding they develop, and independence of the groups. OpenAI notes that the choice of coordinators gives some researchers earlier access to the materials than others.

The mathematical community has responded on three points that bear directly on reporting. First, on *understanding and disclosure*: the Advisory Group on Mathematics and Artificial Intelligence (AGMAI), a group of mathematicians independent of AI companies, distinguishes papers that a human understands, to which the usual norms apply, from results not yet understood by anybody, and asks for the latter that the model, the prompts, a summary of the reasoning, the time and the cost of computation be disclosed, that it be stated how problems were chosen and how many were not solved, and that results be deposited in repositories not controlled by the releasing company (Advisory Group on Mathematics and Artificial Intelligence, 2026). Second, on *formal verification*: a formal check establishes the formal statement, not that it says what the paper says. A human audit of the August release found no confirmed substantive error in a principal result but stressed that Lean "checks the encoded proposition" while human scrutiny must establish its intended meaning, and recommended a standardized report header (Sienicki and Sienicki, 2026); a study of the Navier–Stokes release argues that its formal proof does not correspond to the natural-language proof (Bastounis et al., 2026). The Lean community's own checklist for formal claims asks the same questions: whether the proof is complete, which axioms it depends on, and whether it proves the statement that is claimed (Lean community, 2026). Third, on *records*: press reports noted that only a small fraction of the October manuscripts came with summaries of the model's reasoning, and that natural-language and formal artifacts were not linked by machine-readable metadata (Fernholz, 2026; Howlett and Hall, 2026).

◇ **Discussion** (the prevailing practice and the role of repositories). *In preparing this paper the author observed that most reported uses of AI in our field are of the first kind, a chatbot that helps to establish a lemma and is then acknowledged, and that the second kind needs a fuller account: the form of the panel, the use of a proof assistant, and the place of the human in the loop. The author and the main assistant also noted that this way of working resembles programming more than writing: the proof assistant code, the records of the process and the manuscript source are kept and published together in a version-controlled*

repository, rather than as a single preprint. These observations motivated the items on workflow, formal verification and records in Table 2.

2.2 The approach of this paper compared with large-scale efforts

The case study used a panel of AI agents run by one researcher (Section 8.1). Its design shares several features with the large-scale efforts, and differs from them in scale and in the role of the human. Table 1 summarizes the comparison. OpenAI's descriptions cover the organization of the agents into groups, the variants they received and the consolidation of their findings, but not roles such as reviewer or judge within a group, and the table does not attribute such roles to them.

Feature	**Large-scale efforts (OpenAI, 2026)**	**This paper**
Parallel work on variants	communicating groups prompted with different variants of the problem statement (Navier–Stokes)	proposal agents on distinct angles of one question
Synthesis	a coding agent consolidates the most useful insights of each group into follow-up prompts	judge agents rank the proposals, choose one and graft parts of others onto it
Criticism	model-to-model checking and expert feedback, with human selection among attempts (First Proof); dedicated adversarial reviewers not documented	each proposal and proof attacked by agents instructed to refute it; a review by a model of a second provider
Formal verification	Lean certificates for part of the results	Lean check of the loss-ratio-specific steps of the first formulation, in an abstract setting (Section 8.6)
Release	public repositories with change logs of retractions and repairs	public repository with version history, proof records and process records
Error record	retractions and repairs in a change log	every error found in AI-generated mathematics, with how it was found (Table 5)
Scale	of the order of 10^4 concurrent agents, 88 hours and 2.7 million messages for one problem; thousands of posed problems in the October release	a few to about ten agents per stage; one problem
Configurations	a large communicating run for one problem, with the model updated during the run; a fixed procedure for thousands of problems	one panel design, reused with new roles for the revision; several model versions across sessions (declaration)
Role of the human	choice of problems and of where to direct computation; suggestions of strategies, requests for clarification and selection among attempts (First Proof); external working groups after release	choice of the problem and of every formulation; objections to AI drafts changed the result
Credit and responsibility	no academic credit claimed for discoveries of the models; credit to external groups for the understanding they develop	the author takes responsibility for the result; the division of labour records what the AI contributed
Disclosure of prompts	prompt patterns in an appendix (First Proof); for Navier–Stokes, offered privately to concurrent researchers	the prompts of the latest panel are kept with its records

Table 1: The approach of the case study compared with large-scale multi-agent efforts in mathematics.

The resemblance is in the design principles: several independent attempts from different angles, selection and synthesis of the most useful findings (by a coding agent in the Navier–Stokes run, by judge agents here), criticism, formal verification of what can be formalized, and release of the records with a history of corrections. These principles do not depend on scale, and a single researcher can apply them to a problem in risk management at modest cost. The closest analogue to the case study is OpenAI's First Proof work, in which humans suggested strategies, asked for clarification after expert feedback, used a second model to check the proofs and selected among attempts, which is the role

the author and the review with ChatGPT played here; what the case study adds is a record of the errors found at each step and of the author's interventions. The difference is in the human role. In the large-scale efforts the human chooses problems and allocates computation, and the results may not yet be understood by anybody; in the case study the author directed every stage, and the central idea of the result came from the author's objections to drafts produced by the agents. The assignment of credit differs accordingly: OpenAI claims no academic credit for its models' discoveries and credits external groups with the understanding they develop, whereas here the author, who directed the work and understands the result, takes responsibility for it, and the report records what came from the AI. The paper therefore falls under the first category of the AGMAI guidelines, a result that its author understands, but its proofs were developed by agents, which is why the fuller report of Section 8 is given.

◇ **Discussion** (the design of the panel). *When the author decided to make the use of AI central to the paper, the panel was described to the main assistant as organized in the manner of the recent multi-agent efforts in mathematics, and the paper was conceived as having two parts: the mathematical question, and the demonstration, open to inspection, of how AI was used to answer it. To check this description, two accounts of OpenAI's methods were prepared independently: one by an agent of the main assistant, which read OpenAI's public repositories, the arXiv audits, the AGMAI guidelines and press reports but could not retrieve OpenAI's own pages, and one by ChatGPT, at the author's request, which cited OpenAI's posts directly. The author then retrieved OpenAI's Navier–Stokes and First Proof posts and its October post on academic collaborations in a browser, and the main assistant compared them with both accounts; the posts confirm the facts that both accounts reported, and the text above quotes them directly. The two accounts agree on the facts used here. Both find that the resemblance holds for the principles listed above, that OpenAI used model-to-model checking and coordinated groups of agents, and that no published record specifies dedicated reviewer or judge roles; the comparison in Table 1 is stated accordingly. The use of a second provider to cross-check a description of the literature is an instance of the guardrail of Section 8.4.*

2.3 The proposed standard

We propose that a mathematical paper in risk management or actuarial science whose results were obtained with substantial AI assistance report the items in Table 2, in the tier that fits its use of AI (Section 2.4). For each item the table states whether an editor or reviewer can verify it from public artifacts (V), can inspect it without being able to establish its completeness (P), or must take it on the author's word (S), and where it is reported for the case study.

2.4 Proportionality

A reporting requirement must be proportionate to the use of AI, or it will be ignored or will burden papers that do not need it. We propose three tiers.

(i) *Incidental use*: AI was used for language, literature search, coding unrelated to proofs, or similar tasks, and contributed no mathematical content. Items 1 and 2, which a declaration of AI use covers.

(ii) *Assisted proofs*: AI proposed or checked individual steps, lemmas, examples or counterexamples, which the author verified and which appear in the paper. In addition, items 5, 6 and 8: the errors found in the AI contributions, the checks applied to them, and the verification status of the parts concerned.

(iii) *Agent-developed proofs*: proofs of main results were drafted or developed by AI agents, with the author directing and reviewing. All nine items, with formal verification and a public repository encouraged.

Item	What to report	Tier	Ver.	Case study
1. Tools	providers, models and versions, mode of access, dates of use	1–3	S/P	declaration
2. Division of labour	who posed the question, chose the formulation and decided on content; which tasks were given to AI	1–3	S	Secs. 1, 8.1
3. Workflow	the agents or sessions used, their roles and order, the review loops, and the points at which the author intervened	3	P	Sec. 8.1, 8.7
4. Knowledge base	how the literature was gathered, how reliable each statement was taken to be, which cited statements were checked against the original works	3	P	Sec. 8.2
5. Error log	every error in AI-generated mathematics that was found, how it was found, how it was resolved	2–3	S/P	Table 5
6. Guardrails	the checks applied: adversarial review, review by a second system, recomputation of numbers, counterexamples, self-contained proofs of uncertain citations	2–3	P	Sec. 8.4
7. Formal verification	which results were checked by a proof assistant, in what setting, which steps were taken as hypotheses, and how the formal statements were compared with the paper	3*	V	Sec. 8.6
8. Verification status	for each part of the paper, whether it was reviewed by AI, by a human, or formally checked, and what has not been checked	2–3	S/V	Sec. 8.8
9. Records	the formal proofs, the scripts and the records of the AI process, in a public repository with version history	3*	V	`ai_process/`, `skill/`, `agents/` in the repository

Table 2: Proposed items for reporting AI-assisted proofs. Tier: the tiers in which the item applies (Section 2.4); * encouraged, not required. Ver.: V verifiable from public artifacts, P open to inspection, S self-declared (Section 2.5).

The tier is determined by the most substantial use: a paper whose main theorem was first proved by agents is in tier (iii) even if most of its text was written by the author. The case study is in tier (iii).

2.5 Verifiability and incentives

A journal cannot enforce rules that it cannot check; it can, for instance, not prohibit the use of AI in proofs. The standard therefore asks for information rather than imposing restrictions, and it relies on two mechanisms to make honest reporting the author's best course.

First, items differ in what can be checked. A formal proof can be rebuilt by anyone (V), and a repository with version history shows when each part of the paper and of the formal proof was written and changed (V). Process records and scripts can be inspected and rerun, although their completeness cannot be proved (P). The division of labour and the error log are statements of the author (S). Self-declared items are not unverifiable in isolation: a version history, records and a formal proof constrain them, and an inconsistency between the report and the records is visible to any reader.

Second, the items that cost the author effort are those that save the reviewer effort. A formal check removes the verified steps from the reviewer's work, a verification-status table directs the reviewer to what remains, and an error log shows where AI drafts went wrong and how this was detected. A reviewer who is given these can review an AI-heavy paper faster and with more confidence than one who is told only that AI was used. An author who reports errors found and corrected signals that the result was checked; an author who reports none in an agent-developed paper invites the question of how hard the result was examined. Under these conditions a candid report is in the author's interest.

2.6 Formal verification, repositories and the reviewer

Formal verification in a proof assistant should be encouraged for AI-assisted proofs, but not required. Mathlib does not yet cover all the mathematics used in risk management, and in the case study the classical representation step could not be formalized and entered as a hypothesis (Section 8.6). When a formal check is made, its scope must be reported precisely, as the Navier–Stokes discussion shows: what was formalized, in what setting, which steps were hypotheses, and how the formal statements were compared with the statements of the paper.

A public repository that holds the formal proofs, the scripts and the records of the process, with a version history, should eventually be expected for agent-developed work. Such a repository changes the reviewer's task: part of the verification is done by the proof assistant, and the reviewer's attention shifts to whether the formal statements and definitions say what the paper says, whether the hypotheses are reasonable, and whether the unverified parts are sound. This requires skills that the actuarial research community will have to acquire, notably reading formal statements and rebuilding a formal proof, and it raises the question of how reviewers may themselves use AI. The standard does not settle these questions.

◇ **Discussion** (the editorial perspective). *An editor of an actuarial journal, asked for an opinion on an earlier version of the standard, pointed out that editors are at present more concerned with the volume of submissions and the strain on reviewers than with correctness, that a reporting framework should apply only to AI-heavy work and be proportionate and flexible, that rules which cannot be verified should not be imposed and that a framework should give authors an incentive to report honestly, that formal verification should be encouraged although requiring it is premature, and that repositories should eventually be expected and will change the role of reviewers. In reading these comments with the main assistant, the author identified the changes made in this version: the tiers of Section 2.4, the verifiability column of Table 2 and Section 2.5, the emphasis on the reviewer's work in this subsection, and the restructuring of the paper around the standard, with the mathematical result as its case study and its proofs in the Supplement.*

2.7 Using the standard: recommendations and a reusable skill

The experience of the case study suggests the following recommendations.

For authors who apply the standard. Decide the tier at the start of the work rather than at submission, and set up the repository then. Record errors, discussions and decisions when they occur; reconstructing them afterwards is unreliable. Keep the prompts and outputs of every panel. If a formal check is planned, formalize the definitions early, since this exposes ambiguities in the statements. State the verification status part by part and with dates, so that a later revision of the mathematics does not silently fall outside an earlier check. Name readers or referees whose comments changed the work only with their consent.

For researchers who want to carry out a similar case study. Choose a problem whose answer is not known but whose correctness can be checked, and list its degenerate cases in advance; most of the errors found here occurred at such cases. Give the proposal agents distinct angles, one of which should always be the limits of what can be achieved. Give each proposal its own adversarial checker, and test claims at the points where they can fail rather than at random points (Table 5). Use a model of a second provider for review. Above all, keep the human role active: in the case study the decisive steps came from the author's objections, not from the agents.

For editors and reviewers. Read the verification status first, since it shows where the unverified parts are. Where a formal proof is provided, rebuild it and spend the review effort on whether the formal definitions and statements say what the paper says. Compare the report with the version history of the repository, and check a few entries of the error log against the records.

Practical advice for proofs with agents. The following points summarize what the case study taught about working with agents on proofs; the last three are general experience rather than findings of this

project.

- *Fix the statement before proving it*, with all quantifiers, the domain and the conventions, and ask each agent to restate it; differences between the restatements expose ambiguities.
- *Give every agent the list of degenerate cases* and of established facts; almost every error in Table 5 occurred at a degenerate case, and a list of known facts prevents their rediscovery.
- *Freeze the statement while a proof is checked.* The typical failure is a correct argument for a slightly weaker statement; checkers should compare what was proved with what was claimed.
- *Search for counterexamples before proofs.* The witnesses of Section 7 and Proposition 5.3 arose in this way.
- *Ask for the dependency structure of a proof*, with the hypotheses of each lemma, so that circular citations become visible; treat "clearly" as a gap to be expanded.
- *Require concrete objections*, a counterexample or a failing step, and test on small finite models in exact arithmetic at the points where a claim can fail.
- *Check citations that carry weight* against the source or reprove them; agents misremember hypotheses and numbering.
- *Formalize definitions and statements first*, and have a human compare them with the paper before formal proofs are attempted.
- *Keep the failed approaches*; several became impossibility results of the case study.
- *Ask what a formulation assumes.* The first complete theorem of the case study supplied the premium and the risk measure by assumption, and the author's objection to it produced the central idea of the paper.
- *Give checkers the statement and the proof, not the prover's reasoning*, so that they are not anchored on it; stop checking only when consecutive rounds find nothing new; and keep the working context in files rather than in one long session.

A reusable skill. To make the method usable by others, the protocol of the case study has been written as a *skill*: a structured instruction file that an AI assistant loads when a task matches its description, and that a human can read as a written protocol. It is in the directory `skill/` of the repository. It sets out the roles of author, assistant and agents; the repository layout; the stages of the work (knowledge base, proposal panel, adversarial check, judge, checking runs and repair, second provider, numerical checks, formal verification); the pitfalls met in the case study; templates for the error log, the records of discussions and revisions, and the verification status; the nine items with their tiers; and notes on running the panels in different tools, with the workflow script of the revision as an example. Assistants that support skill files load it directly; with other tools, the roles can be run as separate sessions that share one context document. The skill encodes procedure, not judgment, and it was distilled from one case study; it is offered, like the standard, as a starting point for others to adapt.

Reusable agents. The agent roles of the case study are also provided in a form that others can use, in the directory `agents/` of the repository. Each agent is a text file with a short header, giving its name, a description of when it should be used and the tools it may use, followed by its instructions; this is the format in which Claude Code defines subagents, and with other assistants the instructions can be given to a separate session as its role. The roles are those of Figures 4 and 5 and of the revision panel: a literature agent for one strand, a proposer working from an assigned angle, an adversarial checker instructed to refute one proposal, a judge, a chair who writes the note to the author, a prover, a checking run with a specific brief, a Lean formalizer, and a prompt for an independent review by a model of another provider. All of them read a shared context file, for which a template is given, that fixes the setting, the target, the established facts, the degenerate cases, the author's constraints and the notation; in the case study, a precise context of this kind was the single most effective safeguard. A workflow script runs a complete panel, with proposers in parallel, each followed by its checker, then the judge and the chair. The instructions of each role encode the lessons listed above; for instance, the checker compares the proved statement with the claimed one and tests claims at the points where they can fail. The agent files and the workflow were tested on a problem with a known answer, the characterization

of the functionals $a\max+(1-a)\min$ on a two-point space by cash additivity, comonotonic additivity, monotonicity and law invariance. The panel of six agents recovered the answer with an independence witness for each axiom; an adversarial checker found three presentational errors in one proposal, and the chair marked as a gap the one step that needs the axiom of choice. The test also showed that agent definitions are read when a session starts, so the workflow can instead tell each agent to read its role file directly. The test run is included in `agents/` as a worked example.

◇ **Discussion** (a skill for others). *Towards the end of the work the author asked that the method be written up as a skill file, so that the paper could be used by others, and asked whether such a file can serve someone who wants to carry out a similar case study. The author also asked for the agents designed for the project to be written up in a form that others can use, which led to the agent files described above. The conclusion was that both can serve others, as a protocol, a set of roles and a set of templates: the file is plain text, it is loaded directly by assistants that support skills and can be followed by hand otherwise, and it records the pitfalls that the case study met. It cannot transfer the author's understanding of a problem, which the case study shows to be decisive, and it should be revised as it is used on other problems.*

2.8 Limits of the proposal

The list is a first proposal. It is based on one project, and it will need to be tested on other papers, adapted to results whose proofs are checked formally in full, and revised as the tools change; what AI-assisted research looks like may change within months. It addresses the reporting of results whose authors understand them, and it does not address results produced at scale without human understanding, for which the AGMAI guidelines are more appropriate. Agreement among AI reviews does not establish correctness, and a formal check covers only what was formalized, so the verification status must be stated part by part.

3 The case study: question and related literature

Sections 3 to 7 present the mathematical case study. The question is which maps from losses to $\mathbb{R}\cup\{\infty,\star\}$ are loss ratios ρ/Π of a risk measure and a premium of Choquet type, characterized by axioms on the ratio alone. Three features distinguish it from the classical characterizations of premium principles and risk measures (Yaari, 1987; Wang et al., 1997; Schmeidler, 1986; Wang et al., 2020a). First, the object is a ratio, while those results concern a single functional. Second, the ratio does not determine its numerator and denominator: on positive losses, (ρ,Π) and $(\psi\rho,\psi\Pi)$ give the same ratio for every positive functional ψ. Third, the ratio is not monotone (Proposition 5.21), so the theory of acceptability indices of Cherny and Madan (2009), which rests on monotonicity, does not apply. Table S3 in the Supplement summarizes the differences.

Premia, Choquet integrals and signed Choquet integrals. The axiomatic theory of insurance premia starts from properties of a single functional: translation invariance, no unjustified loading (Gerber, 1974), monotonicity, law invariance and additivity for comonotonic risks. The dual theory of Yaari (1987) leads to the distortion premium $\int g(P(X>x))\,dx$, introduced as a premium principle by Wang (1996) and characterized by Wang et al. (1997); the backbone is the theorem of Schmeidler (1986) that monotone comonotonic additive functionals are Choquet integrals (see also Greco, 1982; Denneberg, 1994). Law-invariant comonotonic coherent risk measures are mixtures of expected shortfalls (Artzner et al., 1999; Delbaen, 2002; Kusuoka, 2001; Jouini et al., 2006; Föllmer and Schied, 2016). Without monotonicity, comonotonic additive functionals of bounded variation are signed Choquet integrals (Murofushi et al., 1994; Cerreia-Vioglio et al., 2012, 2015), and Wang et al. (2020a) prove the law-invariant version used here, together with the characterization of convexity; Wang et al. (2020b) extend it beyond L^∞. All of these results characterize one functional by axioms on that functional. We are not aware of an if-and-only-if characterization of a ratio of two Choquet integrals by axioms on the ratio.

Performance ratios and acceptability indices. Cherny and Madan (2009) characterize monotone, quasi-concave, scale-invariant acceptability indices by increasing families of coherent risk measures, recovered by the cash-shift formula $\rho_x(X) = \inf\{c : \alpha(X+c) \geq x\}$; see also Bielecki et al. (2014); Biagini and Bion-Nadal (2014); Drapeau and Kupper (2013) and, for gain–loss and reward–risk ratios, Bernardo and Ledoit (2000); Biagini and Pinar (2013); Cheridito and Kromer (2013). Han et al. (2025) characterize diversification quotients, the closest if-and-only-if result of level-crossing type. The cash-shift formula is the model for our reading of the premium and the risk measure from cash lines. The differences are of direction, since numerator and denominator of a loss ratio both increase with the loss and the ratio is not monotone; of cardinality, since the loss ratio obeys an accounting identity under pooling; and of object, since we characterize the ratio together with its numerator and denominator.

Implicit functionals and ratio representations. Orlicz premia (Haezendonck and Goovaerts, 1982; Bellini and Rosazza Gianin, 2008), PELVE (Li and Wang, 2023; Fiori and Rosazza Gianin, 2023), return and star-shaped risk measures (Bellini et al., 2018; Castagnoli et al., 2022) are defined implicitly by level crossings; the literature establishes existence and uniqueness but not ratio representations. In decision theory, weighted-utility preferences are represented by a ratio of linear functionals (Chew, 1983), while betweenness yields only implicit representations (Dekel, 1986; Fishburn, 1983); the same distinction separates a weighted pooling identity from a mediant inequality in our setting (Example 7.3). Ratios of expectations are elicitable (Gneiting, 2011), a statistical statement without pooling or cash shifts.

4 The extended loss ratio

4.1 Values and cash lines

Throughout, $(\Omega, \mathscr{F}, P)$ is an atomless probability space and L^∞ is the space of essentially bounded random variables modulo null sets, with the norm $\|\cdot\|_\infty$. Elements of L^∞ represent losses, so positive values are losses and negative values are gains.

Let $\widehat{\mathbb{R}}_\star := \mathbb{R} \cup \{\infty\} \cup \{\star\}$. For reals r, s put

$$[r:s] := \begin{cases} r/s & s \neq 0, \\ \infty & s = 0 \neq r, \\ \star & r = s = 0. \end{cases}$$

The symbol ∞ records "non-zero loss, zero premium" and $\star$ records the indeterminate $0/0$. Only the distinction between finite and non-finite values matters for the theory; the two labels are a convenience (Remark 5.17).

Definition 4.1 (Extended loss ratio, cash line, finite domain)**.** An *extended loss ratio* is a map $\mathrm{LR} : L^\infty \to \widehat{\mathbb{R}}_\star$. Its *cash line* at X is $\phi_X(c) := \mathrm{LR}(X+c)$ for $c \in \mathbb{R}$. Its *finite domain* is $D := \{X : \mathrm{LR}(X) \in \mathbb{R}\}$.

Evaluating LR at cash-shifted positions $X + c$, including negative c, is required by the associated family, since ρ_β is defined through $\mathrm{LR}(X - m) = \beta$. Allowing $c < 0$ is therefore part of the framework rather than an additional assumption. Section S4 treats the actuarial case of nonnegative losses where only $c \geq 0$ is available.

For instance, if $X = \mathbf{1}_A$ with $P(A) = p$, the ratio of the expected loss to the premium $\int_0^\infty \sqrt{P(X > x)}\, dx$ has cash line $c \mapsto (p + c)/(\sqrt{p} + c)$; Example 5.16 and Figure 1 discuss this case.

4.2 Choquet integrals and distortions

For a bounded Borel function $k:[0,1]\to\mathbb{R}$ with $k(0)=0$ define the *(signed) Choquet integral* of a loss $X\in L^\infty$ by

$$I_k(X) := \int_0^\infty k\big(P(X>x)\big)\,dx + \int_{-\infty}^0 \Big[k\big(P(X>x)\big)-k(1)\Big]\,dx.$$

Write $\mathrm{TV}(k)$ for the total variation of k and let

$$\mathscr{H} := \{k:\ \mathrm{TV}(k)<\infty,\ k(0)=0\}, \qquad \mathscr{H}_1 := \{k\in\mathscr{H}:\ k(1)=1\}.$$

A *distortion function* is a non-decreasing element of $\mathscr{H}_1$; jumps are allowed, so value-at-risk-type distortions are included. On nonnegative losses we write $C_k := I_k$. Three facts are used throughout.

- $I_k(X+c)=I_k(X)+c\,k(1)$ (cash slope $k(1)$), and I_k is law invariant.
- For $k\in\mathscr{H}$, I_k is the signed Choquet integral of Wang et al. (2020a, Def. 1), and it is Lipschitz with constant $\mathrm{TV}(k)$ (Wang et al., 2020a).
- I_k is monotone if and only if k is non-decreasing (Wang et al., 2020a), and I_k is convex if and only if k is concave (Wang et al., 2020a, Thm. 3).

Two random variables X,Y are *comonotone* if $(X(\omega)-X(\omega'))(Y(\omega)-Y(\omega'))\geq 0$ for $P\otimes P$-almost every (ω,ω'). Constants are comonotone with everything. A functional F is *comonotonic additive* if $F(X+Y)=F(X)+F(Y)$ whenever X,Y are comonotone. Every I_k with $k\in\mathscr{H}$ is comonotonic additive (proof in Section S1).

5 The axioms, the implied premium and the implied risk measure

In this section we state the axioms and define the premium and the risk measure implied by a loss ratio. Every axiom refers only to values of LR: it evaluates LR at books and at their cash shifts, and uses only arithmetic and whether a value is finite. No axiom refers to a premium, a risk measure, a weight, or a point of a cash line defined from LR. There are five axioms: ($\mathsf{P}_\exists$) and (R) in Section 5.1, (LI) in Section 5.6, and the regularity axioms (RC) and (RM) in Section 5.7. Axioms ($\mathsf{P}_\exists$) and (R) imply three properties, (P), (N) and (CM), which are equivalent to them and which give the axioms their meaning: a unique price point on each cash line, a loss ratio of one for sure losses, and comonotone pooling at relative price (Proposition 5.5). The implied premium and the implied risk measure are minus the price point and minus the zero of the cash line (Definition 5.8 and Corollary 5.14). They are used to interpret the axioms and in the proofs. Table 3 at the end of the section summarizes the axioms and the derived properties.

5.1 Axioms ($\mathsf{P}_\exists$) and (R): pooling of books with zero premium

A premium is the sure amount at which a risk changes hands. A book whose loss ratio is not finite is one that changes hands at zero premium: it has been transferred together with exactly its premium and commands no further premium. We call such a book *unpriced*.

Axiom 5.1 (($\mathsf{P}_\exists$): every book can be made unpriced)**.** For every $X\in L^\infty$ there is $c\in\mathbb{R}$ with $\mathrm{LR}(X+c)\notin\mathbb{R}$.

Axiom 5.2 ((R): comonotone pooling of unpriced books)**.** Let $X,Y\in L^\infty$ be comonotone with $\mathrm{LR}(X)\notin\mathbb{R}$ and $\mathrm{LR}(Y)\notin\mathbb{R}$. Then for all $s,t,u\in\mathbb{R}\setminus\{0\}$ the values $\mathrm{LR}(X+s)$, $\mathrm{LR}(Y+t)$ and $\mathrm{LR}(X+Y+u)$ are real, and

$$u\big(\mathrm{LR}(X+Y+u)-1\big) = s\big(\mathrm{LR}(X+s)-1\big)+t\big(\mathrm{LR}(Y+t)-1\big). \tag{1}$$

Axiom (R) is the accounting identity "premium $\times$ (loss ratio -1) = underwriting result", stated in loss ratios alone. If X is unpriced, adding a sure loss $s \neq 0$ creates a book whose premium is s, since sure amounts are priced at face value, and $s(\mathrm{LR}(X+s)-1)$ is the underwriting result of that book, claims minus premium, in units of cash. Axiom (R) says two things. First, the underwriting result does not depend on the sure amount added, since a sure loss carries no underwriting result: this is (1) with $Y=0$, once it is known that 0 is unpriced and $\mathrm{LR}(t)=1$, both of which (R) implies (Proposition 5.5). Second, the underwriting results of comonotone books, which cannot hedge each other, add up when the books are pooled, and a pool of unpriced books is again unpriced. The hypothesis that X and Y are unpriced is a statement about values of LR; it only selects the books to which the identity applies. A general book enters through its cash shifts. Taking $u=s+t\neq 0$ in (1) gives

$$\mathrm{LR}(X+Y+s+t)=\frac{s}{s+t}\,\mathrm{LR}(X+s)+\frac{t}{s+t}\,\mathrm{LR}(Y+t)\,: \tag{2}$$

two comonotone books written at the premiums s and t pool to the average of their loss ratios weighted by these premiums. This is pooling at relative price, with observed cash premiums as weights.

The hypothesis that X and Y are unpriced cannot be removed, and it cannot be replaced by a condition on finite values of LR. For a loss ratio LR write $\widetilde{\mathrm{LR}}$ for the real-valued map equal to LR where LR is finite and to 1 elsewhere. A condition *depends on* LR *only through its finite values* if it holds for LR whenever it holds for some LR° with $\widetilde{\mathrm{LR}^\circ}=\widetilde{\mathrm{LR}}$. Conditions on limits of finite values along cash lines, such as $\lim_{s\to 0} s(\mathrm{LR}(X+s)-1)$ or $\lim_{|s|\to\infty} s(\mathrm{LR}(X+s)-1)$, are of this kind, since under (P) the values $\mathrm{LR}(X+s)$ near the singular point are finite.

Proposition 5.3 (Non-finite values are needed)**.** *Let* $\Pi'=\max(I_{\sqrt{t}},I_{\min(1,1.5t)})$ *and* $\mathrm{LR}'=[\Pi':\Pi']$. *Then* LR' *satisfies* ($\mathsf{P}_\exists$), (P), (N), (LI), (RC) *and* (RM)*, its finite values are all equal to* 1*, and* LR' *is not of the form* $[I_h:I_g]$ *with* $h,g\in\mathscr{H}_1$*. Consequently, no condition that depends on* LR *only through its finite values and holds for every loss ratio* $[I_h:I_g]$ *can replace* (R) *in Theorems 6.1 and 6.2.*

Proof. Π' is law invariant, monotone, 1-Lipschitz and cash additive with $\Pi'(c)=c$, so the cash line of X under LR' equals 1 except at $c=-\Pi'(X)$, where it is not finite. This gives ($\mathsf{P}_\exists$), (P), (N) and (LI). The sure amount a with $\mathrm{LR}'(X-a)\notin\mathbb{R}$ is $\Pi'(X)$; a book W never has $\mathrm{LR}'(W)=0$, and $\mathrm{LR}'(W)\notin\mathbb{R}$ with $\mathrm{LR}'(W+1)=1$ holds exactly when $\Pi'(W)=0$. Both sure amounts in (RC) and (RM) are therefore $\Pi'(X)$, and both axioms hold. If $\mathrm{LR}'=[I_h:I_g]$, the two maps have the same non-finite books, so $I_g(X)=\Pi'(X)$ for every X by (P), and Π' would be comonotonic additive; Example 7.4 shows that it is not. Finally, $\widetilde{\mathrm{LR}'}\equiv 1=\widetilde{\mathrm{LR}^\circ}$ for $\mathrm{LR}^\circ=[\mathbb{E}:\mathbb{E}]$, which is of the form $[I_h:I_g]$; so every condition of the stated kind holds for LR'. □

On books that break even, the finite values of the cash line carry no information about the premium, and only the location of the non-finite values does (Proposition 5.15). The following remark shows the same for the limit at a vanishing premium.

Remark 5.4 (The limit at zero premium). For $\mathrm{LR}=[I_h:I_g]$, $s(\mathrm{LR}(X+s)-1)=s(I_h(X)-I_g(X))/(I_g(X)+s)$. Its limit as $s\to 0$ exists for every X: it is $I_h(X)$ if X is unpriced and 0 otherwise. An axiom requiring that this limit exist is therefore automatic, and the limit is 0 on unpriced books with $I_h(X)=0$ as well as on all books with a premium, so it does not identify the unpriced books. Nor can (1) be stated for the limits on all comonotone pairs: take $\rho=\mathbb{E}$, $\Pi=I_{\sqrt{t}}$, $X=\mathbf{1}_A$ with $P(A)=\frac{1}{4}$ and $Y=-\frac{1}{2}$. The limits for X and Y are 0, since $\Pi(X)=\frac{1}{2}$ and Y is a sure amount, while $X+Y$ is unpriced and its limit is $\mathbb{E}(X+Y)=-\frac{1}{4}$. The hypothesis of (R) accounts for such pools, whose premiums cancel.

◇ **Discussion** (the hypothesis of (R)). *The author required that no axiom refer to the premium, to a risk measure or even to the price point of a cash line, which is the premium under another name, and that the axioms use the values of* LR *only. Axiom* (R) *was obtained in response (Section 8.7). The author*

then found its hypothesis that $\mathrm{LR}(X)$ *and* $\mathrm{LR}(Y)$ *are not finite an awkward part of an axiom, and asked whether it could be replaced by the requirement that* $\lim_{s\to 0} s\,(\mathrm{LR}(X+s)-1)$ *exist. Remark 5.4 and Proposition 5.3 give the answer worked out with the main assistant: the limit exists for every ratio of the representation and does not single out the books with zero premium, and no condition on finite values can replace* (R)*. The hypothesis was kept, and it is stated through the economic notion of an unpriced book.*

Proposition 5.5 (The derived properties)**.** *Axioms* ($\mathsf{P}_\exists$) *and* (R) *hold if and only if the properties* (P)*,* (N) *and* (CM) *below hold.*

The proof is in Section S1. The rest of this section states the three properties, reads off the implied premium and risk measure, and explains their meaning.

5.2 The price point, the zero and the implied premium

Property 5.6 ((P): one price point on each cash line)**.** For every $X \in L^\infty$ there is exactly one $c \in \mathbb{R}$ at which $\mathrm{LR}(X+c) \notin \mathbb{R}$.

Definition 5.7 (Price point and zero of a cash line)**.** Under (P), the *price point* of X is the unique $c_X^\infty \in \mathbb{R}$ at which $\mathrm{LR}(X+c_X^\infty)$ is not finite. If the cash line of X has at most one zero, its *zero* c_X^0 is the unique $c \in \mathbb{R}$ with $\mathrm{LR}(X+c)=0$ if there is one, and $c_X^0 := c_X^\infty$ if the cash line has no zero.

Both points are read off the cash line: c_X^∞ is where the loss ratio explodes and c_X^0 is where it vanishes. They are the cash offsets at the levels $\beta=\infty$ and $\beta=0$, that is, the sure amounts c with $\mathrm{LR}(X+c)=\beta$; this is the analogue for loss ratios of the recovery $\rho_x(X)=\inf\{c : \alpha(X+c)\geq x\}$ of a family of risk measures from an acceptability index α (Cherny and Madan, 2009). The convention for a cash line without a zero is explained after Proposition 5.13: under the axioms, such a cash line equals 1 away from its price point. Since the cash line of $X+c$ is the cash line of X shifted by c,

$$c_{X+c}^\infty = c_X^\infty - c, \qquad c_{X+c}^0 = c_X^0 - c \qquad \text{for all } X \in L^\infty,\ c \in \mathbb{R}. \tag{3}$$

Definition 5.8 (Implied premium)**.** Under (P), the *implied premium* of X is $\Pi(X) := -c_X^\infty$, minus the price point of X.

By (3),

$$\Pi(X+c) = \Pi(X) + c \qquad \text{for all } X \in L^\infty,\ c \in \mathbb{R}, \tag{4}$$

and X belongs to the finite domain $D=\{\mathrm{LR} \in \mathbb{R}\}$ if and only if $c_X^\infty \neq 0$, that is, $\Pi(X) \neq 0$. Property (P) does not distinguish between the two non-finite values; they play the role of labels (Remark 5.17).

The definition formalizes the notion of a price. A premium is the sure amount at which a risk changes hands: once the insurer holds X and has received $\Pi(X)$, the position $X-\Pi(X)$ commands no further premium, and on the cash line a zero premium is exactly a point where the loss ratio is not finite. In terms of the canonical family $\rho_\beta = (\rho-\beta\Pi)/(1-\beta)$, for books X outside the break-even set, $\rho_\beta(X)$ is the sure offset m at which $\mathrm{LR}(X-m)=\beta$ (Corollary S5.1(b)), and $\rho_\beta \to \Pi$ as $\beta\to\infty$; the implied premium is therefore the member of the associated family at level ∞, in the same way that Cherny and Madan (2009) recover a family of risk measures from an acceptability index α through $\rho_x(X) = \inf\{c : \alpha(X+c)\geq x\}$. The definition also agrees with ratemaking practice, in which the premium is obtained from a loss cost and a permissible loss ratio (Werner and Modlin, 2016). Finally, the price point measures the premium in units of cash, which removes the indeterminacy $(\rho,\Pi)\mapsto(\lambda\rho,\lambda\Pi)$ of the ratio. A shift $X+c$ with $c<0$ is the book transferred together with a sure payment $|c|$.

Property (P) is ordinal: it is unchanged if LR is replaced by $\psi\circ\mathrm{LR}$ for any $\psi:\mathbb{R}\to\mathbb{R}$, the non-finite values being kept. It excludes ratios that are finite along a whole cash line, such as $\arctan(\rho/\Pi)$, and ratios with several singular points.

5.3 Property (N): a sure loss is priced at cost

Property 5.9 ((N))**.** $\mathrm{LR}(c) = 1$ for every constant $c \neq 0$.

A sure loss carries no risk, so its fair premium is its face value; this is the "no unjustified loading" property of premium principles (Gerber, 1974). Under (P), property (N) implies $\Pi(c) = c$ for all c: the cash line of 0 equals 1 away from 0, so $\Pi(0) = 0$, and $\Pi(c) = c$ by (4). Premiums with a proportional loading, such as $(1+\theta)\mathbb{E}$, violate (N).

5.4 Property (CM): comonotone pooling at relative cost

For $X, Y \in D$ with $c_X^\infty + c_Y^\infty \neq 0$ call

$$w_{X,Y} := \frac{c_X^\infty}{c_X^\infty + c_Y^\infty}, \qquad w_{Y,X} = 1 - w_{X,Y} = \frac{c_Y^\infty}{c_X^\infty + c_Y^\infty},$$

the *relative cost* of X and of Y, respectively. Since $-c_X^\infty$ is the sure payment at which X changes hands, $w_{X,Y}$ is the relative price of X in the comonotone portfolio $\{X, Y\}$: the price of X as a fraction of the prices of the two books together, read off their cash lines. In terms of the implied premium, $w_{X,Y} = \Pi(X)/(\Pi(X) + \Pi(Y))$.

◇ **Discussion** (relative price). *In an earlier version the pooling property was written with the premiums inside a single fraction. The author asked that it be written as the combination $w_{X,Y}\mathrm{LR}(X) + w_{Y,X}\mathrm{LR}(Y)$ of the two loss ratios, and that the weight be explained as the relative price of X in the comonotone portfolio $\{X, Y\}$, since this is the form in which an underwriter reads a pooled loss ratio. The paragraph above records that reading.*

Property 5.10 ((CM))**.** For comonotone $X, Y \in D$ with $c_X^\infty + c_Y^\infty \neq 0$,

$$\mathrm{LR}(X+Y) = w_{X,Y}\,\mathrm{LR}(X) + w_{Y,X}\,\mathrm{LR}(Y).$$

Comonotone books do not hedge each other, so the cost of the pooled book should be the sum of the costs of its parts, and then $w_{X,Y}$ is the share of X in the cost of the pool. This is a consequence of (CM) and not part of it: it is the comonotonic additivity $\Pi(X+Y) = \Pi(X) + \Pi(Y)$ of Lemma S1.5. Property (CM) then says that the loss ratio of a comonotone pool is allocated to its constituents according to relative cost: the pooled loss ratio is the combination of the two loss ratios whose weights are their relative costs. Since $\rho = \Pi \cdot \mathrm{LR}$, this is the accounting statement that the loss of the pool is the sum of the losses of the parts and the premium of the pool is the sum of their premiums, so that the loss ratio of the pool is total loss over total premium. For books that are not comonotone, diversification will in general make the pooled premium smaller than the sum of the premiums; this is why Yaari (1987) and Wang et al. (1997) impose additivity only for comonotone risks.

The weights always add up to one. They lie in $[0, 1]$, and the combination is convex, exactly when $\Pi(X)$ and $\Pi(Y)$ have the same sign, in particular when both premiums are positive, which is the case of practical interest. Premiums of opposite sign do occur, since $\Pi(c) = c$ for sure losses and gains, and then one weight is negative and the other exceeds one, so that the combination is affine and not convex.

A constant is comonotone with every book. Taking $Y = c$ in (CM) and using (N) and $\Pi(c) = c$ gives, for $X \in D$, $c \neq 0$ and $c \neq -\Pi(X)$,

$$\mathrm{LR}(X+c) = \frac{\Pi(X)}{\Pi(X)+c}\,\mathrm{LR}(X) + \frac{c}{\Pi(X)+c}\cdot 1 = \frac{\Pi(X)\,\mathrm{LR}(X) + c}{\Pi(X)+c}. \tag{5}$$

Thus along the cash line the loss ratio is the combination of the loss ratio of the book and the loss ratio 1 of the sure loss, with the relative costs of the two as weights: adding a sure loss pulls the ratio towards

1, and the singularity occurs where the relative cost of the book becomes infinite, that is, at $c = -\Pi(X)$. Proposition 5.13 below shows that this makes every cash line a unit-slope Möbius map.

If $\Pi(X)$ and $\Pi(Y)$ have the same sign, then $w_{X,Y} \in [0,1]$ and (CM) makes $\mathrm{LR}(X+Y)$ a convex combination of $\mathrm{LR}(X)$ and $\mathrm{LR}(Y)$; in particular it implies the mediant inequality: $\mathrm{LR}(X+Y)$ lies between $\mathrm{LR}(X)$ and $\mathrm{LR}(Y)$. The inequality is a betweenness property, and, as in decision theory, where betweenness yields only implicit representations while weighted averages yield ratios of linear functionals (Chew, 1983; Dekel, 1986), it does not suffice for our purposes (Example 7.3).

Given (P) and (N), (CM) holds if and only if the implied premium is comonotonic additive and $\Pi \cdot \mathrm{LR}$, restricted to D, extends to a comonotonic additive functional (Lemma S1.5).

Remark 5.11 (A premium given from outside). Suppose that a premium functional Π° is given and (CM) is stated with Π° as weights. If $\mathrm{LR}(X) = a \neq b = \mathrm{LR}(Y)$ and $\mathrm{LR}(X+Y) = L$, then $\Pi^\circ(X)/\Pi^\circ(Y) = (b-L)/(L-a)$, so the weights are determined by the loss ratios. On books with $\mathrm{LR} = 1$, however, the pooled ratio equals 1 whatever the weights, and (CM) carries no information on Π°. For instance, if $\mathrm{LR} \equiv 1$ on $\{I_g \neq 0\}$, then (CM) holds with the non-additive weights $\Pi^\circ = (I_g)^3$, since $a^3 + b^3 \neq 0$ implies $a + b \neq 0$. With the implied premium this difficulty does not arise, because (P) locates the price point also when the ratio equals 1 elsewhere on the cash line.

5.5 The implied risk measure

Definition 5.12 (Implied risk measure)**.** Under (P), put $\rho(X) := \Pi(X)\,\mathrm{LR}(X)$ for $X \in D$.

Proposition 5.13. *Assume* (P), (N) *and* (CM). *Then* ρ *extends uniquely to a cash-additive functional on* L^∞, *with* $\rho(c) = c$, *and for every* $X \in L^\infty$:

(a) $\mathrm{LR}(X+c) = \dfrac{\rho(X)+c}{\Pi(X)+c}$ *for every* $c \neq -\Pi(X)$;
(b) $\rho(X) = (\Pi(X)+c)\,\mathrm{LR}(X+c) - c$ *for every* $c \neq -\Pi(X)$;
(c) $\rho(X) - \Pi(X) = \lim_{c \to -\Pi(X)} (c + \Pi(X))\,\mathrm{LR}(X+c)$; *in particular* $\rho(X) = \lim_{c \to 0} c\,\mathrm{LR}(X+c)$ *if* $\Pi(X) = 0$;
(d) if $\rho(X) \neq \Pi(X)$, *then* $-\rho(X)$ *is the unique zero of the cash line of* X; *if* $\rho(X) = \Pi(X)$, *the cash line equals* 1 *away from the price point.*

The proof is given in Section S1. By (c), the implied risk measure of a book with zero premium is the residue of its cash line at the price point: if a small premium c is introduced, $\mathrm{LR}(X+c) \approx \rho(X)/c$. We call $N := \{\rho = \Pi\}$ the *break-even set*; $N \cap D = \{\mathrm{LR} = 1\}$, and N also contains the books with $\rho = \Pi = 0$. On N the residue vanishes and, by (d), the cash line has no zero, so that the risk measure cannot be read from the zero of the cash line; this is why it is defined through (b).

Corollary 5.14 (The implied risk measure is minus the zero)**.** *Assume* (P), (N) *and* (CM). *Then every cash line has at most one zero, and* $\rho(X) = -c_X^0$ *for every* $X \in L^\infty$.

Proof. By Proposition 5.13(d), if $\rho(X) \neq \Pi(X)$ the cash line of X has the unique zero $-\rho(X)$. If $\rho(X) = \Pi(X)$, it equals 1 away from the price point and has no zero, so $c_X^0 = c_X^\infty = -\Pi(X) = -\rho(X)$. □

Thus the implied risk measure, like the implied premium, is a point of the cash line: $\rho(X)$ is minus the sure amount at which the loss ratio of X vanishes, and on the break-even set the convention of Definition 5.7 makes the two points coincide.

Away from the break-even set, the premium can be computed explicitly from two values of the loss ratio. Solving (5) for $\Pi(X)$ gives the following formulas.

Proposition 5.15 (Explicit formulas)**.** *Assume* (P), (N) *and* (CM), *and let* $X \in D$ *with* $\mathrm{LR}(X) \neq 1$. *Write* $\phi_X(c) = \mathrm{LR}(X+c)$.

(a) Two quotes. *For every* $c \notin \{0, -\Pi(X)\}$, $\phi_X(c) \neq \mathrm{LR}(X)$ *and*

$$\Pi(X) = \frac{c\,(1-\phi_X(c))}{\phi_X(c)-\mathrm{LR}(X)}, \qquad \rho(X) = \mathrm{LR}(X)\,\frac{c\,(1-\phi_X(c))}{\phi_X(c)-\mathrm{LR}(X)}.$$

(b) Sensitivity to a sure loss. ϕ_X *is differentiable at* 0, $\phi_X'(0) = (1-\mathrm{LR}(X))/\Pi(X) \neq 0$, *and*

$$\Pi(X) = \frac{1-\mathrm{LR}(X)}{\phi_X'(0)}, \qquad \rho(X) = \frac{\mathrm{LR}(X)\,(1-\mathrm{LR}(X))}{\phi_X'(0)}.$$

(c) Underwriting result. $\Pi(X) - \rho(X) = \lim_{|c|\to\infty} c\,(1-\phi_X(c))$.

If $\mathrm{LR}(X) = 1$, *then* $\phi_X \equiv 1$ *away from the price point, and* $\Pi(X)$ *is not determined by the finite values of the cash line.*

Proof. By Proposition 5.13(a), $\phi_X(c) = (\rho + c)/(\Pi + c)$ with $\rho = \rho(X)$, $\Pi = \Pi(X) \neq 0$, so

$$1 - \phi_X(c) = \frac{\Pi - \rho}{\Pi + c}, \qquad \phi_X(c) - \mathrm{LR}(X) = \frac{c\,(\Pi-\rho)}{\Pi\,(\Pi+c)}.$$

Since $\mathrm{LR}(X) \neq 1$ means $\rho \neq \Pi$, the second expression is non-zero for $c \neq 0$, and the quotient gives (a). Differentiating, $\phi_X'(c) = (\Pi-\rho)/(\Pi+c)^2$, so $\phi_X'(0) = (1-\mathrm{LR}(X))/\Pi$, which gives (b); and $c\,(1-\phi_X(c)) = c\,(\Pi-\rho)/(\Pi+c) \to \Pi - \rho$, which gives (c). If $\mathrm{LR}(X) = 1$, then $\rho = \Pi$ and $\phi_X(c) = 1$ for $c \neq -\Pi$. □

Formula (b) reads: the implied premium is the underwriting margin $1 - \mathrm{LR}(X)$ divided by the sensitivity of the loss ratio to a sure loss; formula (a) is its finite-difference version, which needs only two quoted loss ratios, with and without a sure loss c. For $X = \mathbf{1}_A$ in Example 5.16, $\mathrm{LR}(X) = \frac{1}{2}$ and $\phi_X(1) = \frac{5}{6}$, so (a) gives $\Pi(X) = \frac{1}{6}/\frac{1}{3} = \frac{1}{2}$. For books outside D the formulas apply to $X + c$ with $X + c \in D$, and $\Pi(X) = \Pi(X+c) - c$ by (4).

The last statement of Proposition 5.15 is the reason why the axioms must refer to non-finite values of LR, as ($\mathsf{P}_\exists$) and (R) do. On the break-even set the cash line carries no information about the premium except the location of its singularity, and Remark 5.11 shows that the weights in (CM) are then not identified by the values of the ratio. Off the break-even set, the weights of (CM) could equally be written with the explicit premium of (a) or (b) in place of $-c^\infty$; but (CM) must also govern pools that contain break-even books, and there only the price point locates the premium. Axiom (R) avoids weights altogether: it moves each book to zero premium by a sure payment, which is a statement about non-finite values, and pools there.

◇ **Discussion** (explicit premium formulas). *After the remark of Peng Liu reported in Section 8.7, the author asked whether the implicitly defined premium could be replaced by an explicit formula in the values of* LR, *so that the axioms could be restated in terms of the loss ratio. Three proposals were examined. The first solves the identity (5) for* $\Pi(X)$, *which gives (a) and (b) of Proposition 5.15. The second, proposed by the author, takes the existence of* $M(X) := \lim_{|c|\to\infty} c\,(1-\mathrm{LR}(X+c))$ *as a condition and defines the premium as* $M(X)/(1-\mathrm{LR}(X))$; *by (c) it agrees with the implied premium off the break-even set. The third, also proposed by the author, treats a break-even book* X *by perturbation, setting* $\Pi(X) := \lim_{\varepsilon\to 0} \Pi(X+\varepsilon Y)$ *for a book* Y *such that* $X + \varepsilon Y$ *is outside the break-even set, with* $\Pi(X+\varepsilon Y)$ *given by the second formula. The main assistant checked that each proposal agrees with the implied premium wherever it is defined, and that the perturbation recovers it on the break-even set provided the premium is already known to be continuous, which is part of what the axioms are meant to deliver. The conclusion was that no explicit formula can serve as a definition on its own: on break-even books, and for* $h = g$ *on every book, the finite values of the cash line carry no information about the premium. This led to the formulation of Section 5.1, in which the premium enters only through the non-finite values of* LR.

◇ **Discussion** (the margin ratio). *The author also asked how the cash-line formulation of an earlier version (Remark S2.2 in the Supplement) could be written with the loss ratio alone. Under the representation,* $1-\mathrm{LR}(X+c)=(\Pi(X)-\rho(X))/(\Pi(X)+c)$*, so off the break-even set the map* $c\mapsto 1/(1-\mathrm{LR}(X+c))$ *is affine; it vanishes at the price point and equals* 1 *where the cash line vanishes. The author proposed to call* $\mathrm{MR}:=1-\mathrm{LR}$ *the* margin ratio*, the underwriting margin per unit of premium, and to state the condition through the ratio of the two: since* $1/(1-\mathrm{LR})=1+\mathrm{LR}/\mathrm{MR}$*, every cash line off the break-even set is a unit-slope Möbius map if and only if* $c\mapsto \mathrm{LR}(X+c)/\mathrm{MR}(X+c)$ *is affine, with slope* $1/(\Pi(X)-\rho(X))$*. The author's first version of the identity had the quotient inverted, and the main assistant corrected it. Both forms degenerate on the break-even set, where the margin ratio vanishes along the whole cash line except at the price point, so they do not locate the premium of a break-even book. The condition was therefore kept as a reading of Proposition 5.13(a), not as an axiom.*

Example 5.16 (Three cash lines)**.** Let $\rho=I_{\mathrm{id}}=\mathbb{E}$ and $\Pi=I_{\sqrt{\cdot}}$, and let $P(A)=\frac{1}{4}$. Figure 1 shows the cash lines of three books.

(i) $X=\mathbf{1}_A$: $\mathrm{LR}(X+c)=(\frac{1}{4}+c)/(\frac{1}{2}+c)$. The price point is $-\frac{1}{2}$, so $\Pi(X)=\frac{1}{2}$, and $\rho(X)=\Pi(X)\mathrm{LR}(X)=\frac{1}{4}$; the cash line vanishes at $-\frac{1}{4}$.

(ii) $Y=\mathbf{1}_A-\frac{1}{2}$, a book with zero premium: $\mathrm{LR}(Y)=\infty$ and $\mathrm{LR}(Y+c)=(c-\frac{1}{4})/c$. The residue at the price point 0 is $-\frac{1}{4}=\rho(Y)$.

(iii) $Z\equiv\frac{1}{2}$, a break-even book: $\mathrm{LR}(Z+c)=1$ for $c\neq-\frac{1}{2}$, and $\mathrm{LR}(Z-\frac{1}{2})=\star$. The price point gives $\Pi(Z)=\frac{1}{2}$; the cash line has no zero, and $\rho(Z)=\Pi(Z)\cdot 1=\frac{1}{2}$.

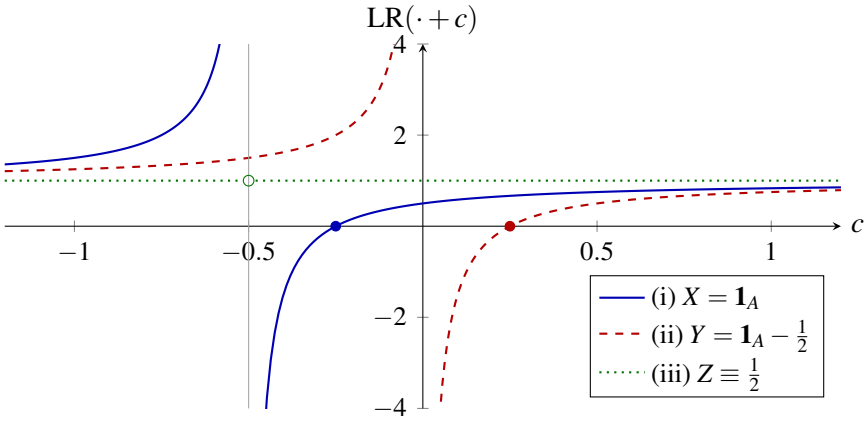


Figure 1: Cash lines of the three books of Example 5.16. Book (i) explodes at $c=-\frac{1}{2}$ (its price point) and vanishes at $c=-\frac{1}{4}$; book (ii) explodes at $c=0$ with residue $-\frac{1}{4}$; book (iii) equals 1 except at its price point $c=-\frac{1}{2}$, where the singularity is removable (open circle).

Remark 5.17 (Labels at the price point). To have $\mathrm{LR}=[\rho:\Pi]$ also at price points, one may adopt the convention

(S) at the price point, $\mathrm{LR}=\star$ if the residue in (c) is zero and $\mathrm{LR}=\infty$ otherwise.

Convention (S) is not used in the proofs. If, instead, the loss ratio is real valued with $0/0:=1$, the price point of a break-even book cannot be observed and the implied premium is not defined there.

5.6 Axiom (LI): law invariance

Axiom 5.18 ((LI))**.** If $X\overset{d}{=}Y$, then either $\mathrm{LR}(X)=\mathrm{LR}(Y)$ or both values are non-finite.

This is the standard assumption of premium and risk-measure theory (Wang et al., 1997; Kusuoka, 2001). Because the implied premium and risk measure are computed from cash lines, (LI) passes to them (Lemma S1.6). Ratios that weight scenarios non-uniformly, such as $\mathbb{E}_Q/\mathbb{E}$ with $Q\neq P$, violate (LI); they are covered by Theorem S3.1.

5.7 Axioms (RC) and (RM): regularity of the cash offsets

The regularity axioms concern two sure amounts attached to a book: the amount a whose payment leaves the book unpriced, $\mathrm{LR}(X-a) \notin \mathbb{R}$, and the amount a whose payment leaves it without loss. A book W has no loss if $\mathrm{LR}(W) = 0$, or, when W is unpriced, if $\mathrm{LR}(W+1) = 1$, that is, if W written at the premium 1 costs exactly that premium.

Axiom 5.19 ((RC))**.** For every $\varepsilon > 0$ there is $\delta > 0$ such that for all $X, Y \in L^\infty$ with $\|X - Y\|_\infty < \delta$ and all $a, b \in \mathbb{R}$:

(i) if $\mathrm{LR}(X-a) \notin \mathbb{R}$ and $\mathrm{LR}(Y-b) \notin \mathbb{R}$, then $|a-b| < \varepsilon$;
(ii) if each $W \in \{X-a, Y-b\}$ satisfies [$\mathrm{LR}(W) = 0$, or $\mathrm{LR}(W) \notin \mathbb{R}$ and $\mathrm{LR}(W+1) = 1$], then $|a-b| < \varepsilon$.

Axiom 5.20 ((RM))**.** Let $X \le Y$ almost surely and $a, b \in \mathbb{R}$.

(i) If $\mathrm{LR}(X-a) \notin \mathbb{R}$ and $\mathrm{LR}(Y-b) \notin \mathbb{R}$, then $a \le b$.
(ii) If each $W \in \{X-a, Y-b\}$ satisfies [$\mathrm{LR}(W) = 0$, or $\mathrm{LR}(W) \notin \mathbb{R}$ and $\mathrm{LR}(W+1) = 1$], then $a \le b$.

Taking $X = Y$ shows that each axiom forces the two amounts to be unique, so the axioms make sense independently of the others. Under ($\mathsf{P}_\exists$) and (R), the amount in (i) is the implied premium $\Pi(X)$ and the amount in (ii) is the implied risk measure $\rho(X)$ (Lemma S1.2); thus (RC) (respectively (RM)) says exactly that Π and ρ are uniformly continuous (monotone) on L^∞, and (RM) implies (RC) (Lemma S1.7). In terms of the cash lines, the two amounts are minus the price point and minus the zero. Axiom (RM) says that a larger loss must be offset by a larger sure amount, both to be priced at zero (level ∞) and to have no loss (level 0). This is the monotonicity of the cash offsets familiar from acceptability indices (Cherny and Madan, 2009), imposed at the two levels that determine the cash line; (RC) is the corresponding continuity. For a ratio $[A:B]$ with A, B cash additive, the two amounts are $B(X)$ and $A(X)$, so that the axioms reduce to continuity or monotonicity of A and B. We include the signed tier (RC) because the canonical family ρ_β is a difference of monotone functionals and is in general not monotone (Corollary S5.1 and Example S5.2). Monotonicity is imposed on the cash offsets and not on the values of the ratio, for the following reason.

Proposition 5.21 (No monotone loss ratio)**.** *Let $h, g \in \mathscr{H}_1$ with $g > 0$ on $(0,1]$, and let $\mathrm{LR} = I_h / I_g$ on $\mathscr{X} := L^\infty_+ \setminus \{0\}$. If LR is monotone on $\mathscr{X}$ (non-decreasing or non-increasing for the almost sure order), then $h = g$ and $\mathrm{LR} \equiv 1$.*

Proof. Let $P(A) = p > 0$. For $c \ge 0$, $\mathrm{LR}(\mathbf{1}_A + c) = \frac{h(p)+c}{g(p)+c}$ is monotone in c with derivative at 0 of the sign of $g(p) - h(p)$. Next let $P(E) = q > 0$ and $t \ge 0$. Then $X_t = 1 + t\mathbf{1}_E \in \mathscr{X}$ is increasing in t, and $I_k(X_t) = 1 + t\,k(q)$ for $k \in \{h, g\}$, so $\mathrm{LR}(X_t) = \frac{1+t\,h(q)}{1+t\,g(q)}$ has derivative at $t = 0$ of the sign of $h(q) - g(q)$. If LR is non-decreasing both signs are non-negative, hence $h = g$ on $(0,1]$; if LR is non-increasing both are non-positive, and again $h = g$. □

Consequently, the theorems on acceptability indices (Cherny and Madan, 2009), which require monotonicity of the index, do not apply to loss ratios. Axiom (RM) is nevertheless an order property of LR: it concerns the sure amounts at which the ratio becomes non-finite or vanishes, not the size of the ratio.

5.8 Summary

6 Main results

Throughout this section $\mathrm{LR} : L^\infty \to \widehat{\mathbb{R}}_\star$, and Π and ρ denote the implied premium and the implied risk measure (Definitions 5.8 and 5.12, Proposition 5.13).

Axiom	Statement	Interpretation and counterpart
($\mathsf{P}_\exists$)	each cash line has a non-finite value	every book can be made unpriced by a sure payment
(R)	comonotone pooling of unpriced books, (1)	underwriting results add; comonotonic additivity (Yaari, 1987; Wang et al., 1997)
(LI)	law invariance	standard (Wang et al., 1997; Kusuoka, 2001)
(RC)/(RM)	regularity of the sure amounts that make a book unpriced and loss-free	continuity / monotonicity of premium and risk measure; cash offsets (Cherny and Madan, 2009)
Derived properties, equivalent to ($\mathsf{P}_\exists$) *and* (R) *(Proposition 5.5)*		
(P)	one price point on each cash line	defines the premium; the level $\beta = \infty$ of the associated family
(N)	$\mathrm{LR}(c) = 1$ for $c \neq 0$	no unjustified loading (Gerber, 1974)
(CM)	comonotone pooling, weighted by relative price read off the price points	premium-weighted average

Table 3: The axioms and the derived properties. Every axiom refers only to values of LR.

6.1 The two characterization theorems

Theorem 6.1 (Signed tier)**.** *The following are equivalent.*

(i) LR *satisfies* ($\mathsf{P}_\exists$), (R), (LI) *and* (RC).

(ii) There is a pair $(h,g) \in \mathscr{H}_1^2$ *such that* $\mathrm{LR}(X) = I_h(X)/I_g(X)$ *whenever* $I_g(X) \neq 0$*, and* $\mathrm{LR}(X) \notin \mathbb{R}$ *whenever* $I_g(X) = 0$*.*

In that case the price point and the zero of the cash line of X *are* $-I_g(X)$ *and* $-I_h(X)$*; that is, the implied premium is* $\Pi = I_g$ *and the implied risk measure is* $\rho = I_h$*. If, in addition, the labelling convention* (S) *holds, then* $\mathrm{LR} = [I_h : I_g]$ *on all of* L^∞*.*

Theorem 6.2 (Distortion tier)**.** *The following are equivalent.*

(i) LR *satisfies* ($\mathsf{P}_\exists$), (R), (LI) *and* (RM).

(ii) Statement (ii) of Theorem 6.1 holds with h,g *distortion functions (non-decreasing elements of* $\mathscr{H}_1$*).*

By Proposition 5.5, condition (i) may equivalently be stated with the properties (P), (N) and (CM) in place of the axioms ($\mathsf{P}_\exists$) and (R). Section 7 shows that none of the axioms can be dropped. The version without (LI), with a representation by set functions, is Theorem S3.1 in the Supplement.

The proofs are given in Section S1 of the Supplement. The argument runs as follows. (1) Applied to a book with zero premium and itself, (R) shows that the non-finite point of each cash line is unique, which defines the implied premium Π; applied to sure losses it gives $\mathrm{LR}(c) = 1$. (2) Applied to an unpriced book W and the sure loss 0, it shows that $r(W) = s\,(\mathrm{LR}(W+s) - 1)$ does not depend on s; with $\rho(X) := \Pi(X) + r(X - \Pi(X))$, every cash line is the unit-slope Möbius map $c \mapsto (\rho(X)+c)/(\Pi(X)+c)$. (3) Applied to the unpriced books $X - \Pi(X)$ and $Y - \Pi(Y)$, (R) gives $\Pi(X+Y) = \Pi(X) + \Pi(Y)$ and $\rho(X+Y) = \rho(X) + \rho(Y)$ for all comonotone X, Y, including books that break even, on which no finite value of LR identifies the premium. (4) The sure amounts in (RC) and (RM) are $\Pi(X)$ and $\rho(X)$, so these axioms are continuity and monotonicity of Π and ρ, and (LI) passes to both. (5) A law-invariant, comonotonic additive functional with $F(c) = c$ that is uniformly continuous (respectively monotone) is a signed Choquet integral (respectively a distortion premium) (Wang et al., 2020a, 1997); this step is reproved in the Supplement. The converse follows from comonotonic additivity and Lipschitz continuity of I_h and I_g.

6.2 Uniqueness and derived properties

Proposition 6.3. *In Theorems 6.1 and 6.2, (h,g) is unique. For an event A with $P(A)=p$,*

$$g(p) = -c^{\infty}_{\mathbf{1}_A}, \qquad c^{\infty}_{\mathbf{1}_A} = \text{the price point of } \mathbf{1}_A;$$
$$h(p) = \big(g(p)+c\big)\,\mathrm{LR}(\mathbf{1}_A + c) - c \qquad \text{for any } c \neq -g(p).$$

Moreover $D = \{I_g \neq 0\}$ and the break-even set is $N := \{\rho = \Pi\} = \{I_{h-g} = 0\}$. Within Theorem 6.1, ρ (respectively Π) is monotone if and only if h (respectively g) is non-decreasing.

Proposition 6.4. *Under the axioms of Theorem 6.1:*

(i) every cash line is a unit-slope Möbius map, $\mathrm{LR}(X+c) = (\rho(X)+c)/(\Pi(X)+c)$ *for* $c \neq -\Pi(X)$*;*
(ii) ρ and Π are cash additive, $\rho(c) = \Pi(c) = c$, and they are Lipschitz with constants $\mathrm{TV}(h)$ *and* $\mathrm{TV}(g)$*;*
(iii) ρ and Π are comonotonic additive and positively homogeneous, hence LR *is scale invariant:* $\mathrm{LR}(\lambda X) = \mathrm{LR}(X)$ *for $\lambda > 0$ and $X \in D$, and for all X under* (S).

Properties that are often postulated for the numerator and the denominator of a loss ratio are thus derived; the association itself is derived in Corollary S5.1.

6.3 Examples

Example 6.5 (Expected loss over a proportional hazard premium)**.** Let $h = \mathrm{id}$ and $g_r(p) = p^r$ with $r \in (0,1)$, so that $\rho = \mathbb{E}$ and Π is the proportional hazard premium. For $P(A) = p$, $\mathrm{LR}(\mathbf{1}_A) = p^{1-r}$, and Proposition 6.3 recovers the premium distortion from the loss ratios of Bernoulli books, $g_r(p) = p/\mathrm{LR}(\mathbf{1}_A)$. For a loss U uniform on $[0,1]$, $\mathbb{E}U = \frac{1}{2}$ and $\Pi(U) = \int_0^1 (1-x)^r\,dx = 1/(1+r)$, so $\mathrm{LR}(U) = (1+r)/2$. Since g_r is concave and h is linear, $\mathrm{LR} \leq 1$ on $\mathscr{X}$ (Corollary S5.4). The left panel of Figure 2 shows $p \mapsto \mathrm{LR}(\mathbf{1}_A)$ for three values of r.

Example 6.6 (Expected shortfall over expected loss)**.** Let $h_\alpha(p) = \min\{p/(1-\alpha), 1\}$, $\alpha \in (0,1)$, and $g = \mathrm{id}$, so that $\rho = \mathrm{ES}_\alpha$ and $\Pi = \mathbb{E}$. Then $\mathrm{LR}(\mathbf{1}_A) = \min\{1/(1-\alpha), 1/p\}$ and $\mathrm{LR}(U) = 1+\alpha$ for U uniform on $[0,1]$. Here h_α is concave and g is convex, so $\mathrm{LR} \geq 1$ and LR is quasi-convex on $\{\Pi > 0\}$ (Corollary S5.4(d)); see the right panel of Figure 2.

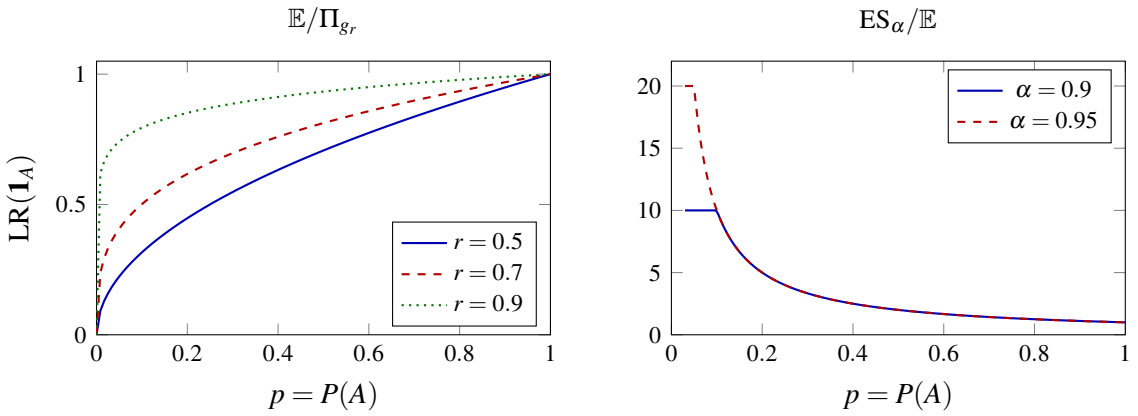


Figure 2: Loss ratios of Bernoulli books $\mathbf{1}_A$ as functions of $p = P(A)$. Left: expected loss over the proportional hazard premium (Example 6.5); the curves identify $g_r(p) = p/\mathrm{LR}(\mathbf{1}_A)$. Right: expected shortfall over expected loss (Example 6.6).

7 Independence of the axioms

For each of the axioms ($\mathsf{P}_\exists$), (R), (LI), (RC) and (RM) we give a loss ratio that satisfies the remaining axioms and is not of the form in Theorem 6.1 or 6.2. Two facts shorten the verifications. First, by Proposition 5.5, a ratio satisfies ($\mathsf{P}_\exists$) and (R) exactly when it has the properties (P), (N) and (CM). Second, for a ratio $[A:B]$ of cash-additive functionals the sure amounts that make X unpriced and loss-free are $B(X)$ and $A(X)$ (Section 5.7), so (RC) and (RM) reduce to continuity and monotonicity of A and B. Table 4 summarizes the examples.

Dropped	**Witness**	**Why it is a witness**
($\mathsf{P}_\exists$)	$\mathrm{LR} \equiv 1$	No value is non-finite; (R), (LI), (RC), (RM) hold vacuously.
(R)	$[\max(I_{\sqrt{t}}, I_{\min(1,1.5t)}) : \mathbb{E}]$; also $\psi \circ [I_h : I_g]$, $\psi(t) = t^3$ (Example 7.3), and $[\Pi' : \Pi']$	Implied risk measure (resp. premium) is cash additive but not comonotonic additive; the cube has poles of order three.
(LI)	$[\mathbb{E}_Q : \mathbb{E}]$, $Q \neq P$ with bounded density	All other axioms hold; $\mathbb{E}_Q$ is not law invariant.
(RC) (Thm. 6.1)	$[I_h : \mathbb{E}]$ with h continuous, not of bounded variation	Uniform continuity fails; h is not of bounded variation.
(RM) (Thm. 6.2)	$[I_h : \mathbb{E}]$, $h(t) = t + \frac{1}{4}\sin 2\pi t$	h is not monotone, so no distortion pair represents it.

Table 4: Witnesses for the axioms of Theorems 6.1 and 6.2. Here $\Pi' = \max(I_{\sqrt{t}}, I_{\min(1,1.5t)})$.

Example 7.1 (Independence of ($\mathsf{P}_\exists$))**.** The constant ratio $\mathrm{LR} \equiv 1$ takes no non-finite value, so ($\mathsf{P}_\exists$) fails. Axioms (R), (RC) and (RM) hold vacuously, since their hypotheses require a non-finite value or a zero, and (LI) is clear. It is not of the form in Theorem 6.1: every such ratio is non-finite at $X = 0$, since $I_g(0) = 0$.

Example 7.2 (A loading on sure losses violates (R))**.** Fix $\kappa > 0$, $\kappa \neq 1$, and let $\mathrm{LR} = [I_h : \kappa I_g]$ with distortions h, g. The cash line of X is $(I_h(X) + c)/(\kappa(I_g(X) + c))$, non-finite only at $c = -I_g(X)$, so ($\mathsf{P}_\exists$) and (P) hold with $\Pi = I_g$, and with these weights the pooling identity (CM) holds because I_h and I_g are comonotonic additive. But $\mathrm{LR}(c) = 1/\kappa \neq 1$, so (N) fails and, by Proposition 5.5, so does (R). Directly: for unpriced X, $s(\mathrm{LR}(X+s) - 1) = (I_h(X) + s(1-\kappa))/\kappa$ depends on s. The normalization on sure losses is therefore built into (R): it is what excludes loadings on sure losses. (We conjecture, but have not proved, that an analogue of Theorem 6.1 with a known loading κ holds with $\mathrm{LR} = [I_h : \kappa I_g]$.)

Example 7.3 (Ordinal information is not enough)**.** Let $h \neq g$ be distortions and $\mathrm{LR}' = \psi \circ [I_h : I_g]$ with $\psi(t) = t^3$ and ψ fixing the non-finite values. Since ψ maps $\mathbb{R}$ to $\mathbb{R}$, LR' has the same price points as $[I_h : I_g]$, so ($\mathsf{P}_\exists$) and (P) hold with $\Pi = I_g$; (N) holds because $1^3 = 1$; (LI) holds, and so does the mediant inequality, because ψ is increasing. But LR' is not of the form in Theorem 6.1: at a book X with $I_h(X) \neq I_g(X)$ (for instance an indicator at a point where $h \neq g$), the cash line $\big((I_h(X)+c)/(I_g(X)+c)\big)^3$ has a pole of order three, while every cash line in Theorem 6.1 has a simple pole. Hence (CM) fails, and so does (R) by Proposition 5.5. Since ψ maps 0 to 0, 1 to 1 and only 1 to 1, and keeps the non-finite values, the sure amounts that make a book unpriced or loss-free are the same for LR' as for $[I_h : I_g]$, namely $I_g(X)$ and $I_h(X)$; so (RC) and (RM) hold as well, and the example satisfies every axiom except (R). Betweenness-type, ordinal information determines the loss ratio only up to an increasing transformation, as for Chew (1983) and Dekel (1986); the cardinal axiom (R) cannot be avoided.

Example 7.4 (Independence of (R))**.** Let $\mathrm{LR} = [\rho : \mathbb{E}]$ with $\rho = \max(I_{\sqrt{t}}, I_{\min(1,1.5t)})$. ($\mathsf{P}_\exists$) and (P) hold with $\Pi = \mathbb{E}$, (N) holds, and the implied risk measure is ρ, a maximum of cash-additive maps; (LI), (RM) and (RC) hold. Take $E \subset B$ with $P(B) = 0.5$, $P(E) = 0.1$ and $X = \mathbf{1}_B + \mathbf{1}_E$. Then $\rho(\mathbf{1}_B) = \max(0.707, 0.75) = 0.75$ and $\rho(\mathbf{1}_E) = \max(0.316, 0.15) = 0.316$, with sum 1.066, while

$\rho(X) = \max(0.707 + 0.316,\ 0.75 + 0.15) = 1.023$. So ρ is not comonotonic additive and (CM) fails by Lemma S1.5(a); hence (R) fails.

The ratio $[\Pi' : \Pi']$ with $\Pi' = \rho$ shows more: it equals 1 wherever it is finite, and satisfies ($\mathsf{P}_\exists$), (LI), (RC) and (RM) (both sure amounts equal $\Pi'(X)$, which is monotone and 1-Lipschitz). It fails (R) only through the clause that a pool of unpriced books is unpriced: for the books above, $\mathbf{1}_B - \Pi'(\mathbf{1}_B)$ and $\mathbf{1}_E - \Pi'(\mathbf{1}_E)$ are unpriced but their sum is not: its premium is $1.023 - 1.066 = -0.043$. Hence no axiom that depends on LR only through its finite values can replace (R) (Proposition 5.3).

Example 7.5 (Independence of (LI))**.** Let $Z = dQ/dP$ be bounded, non-constant, $Z \geq 0$ and $\mathbb{E}Z = 1$, and $\mathrm{LR} = [\mathbb{E}_Q : \mathbb{E}]$. (P) holds with $\Pi = \mathbb{E}$, (N) holds, (CM) holds by Lemma S1.5(b) because $\mathbb{E}_Q$ and $\mathbb{E}$ are additive, so ($\mathsf{P}_\exists$) and (R) hold, and (RM), (RC) hold. But $\mathbb{E}_Q(\mathbf{1}_A) = Q(A)$ is not a function of $P(A)$, so the implied risk measure is not of the form I_h. Without (LI) the class is exactly that of Theorem S3.1 in the Supplement.

Example 7.6 (Independence of (RC) in Theorem 6.1)**.** Let $h(t) = t\big(1 + \frac{1}{2}\sin(1/t)\big)/\big(1 + \frac{1}{2}\sin 1\big)$, continuous with $h(0) = 0$, $h(1) = 1$, and not of bounded variation (Figure 3) ($\int |h'|$ diverges at 0). Let $\mathrm{LR} = [I_h : \mathbb{E}]$. (P), (N) and (LI) are clear; I_h is norm continuous (survival functions converge off a countable set, and dominated convergence applies) and additive on simple comonotone positions by the layer computation, hence comonotonic additive, so (CM) holds, and with it ($\mathsf{P}_\exists$) and (R). The implied risk measure is I_h, and by Lemma S1.9 (RC) would force h to be of bounded variation. So (RC) fails and no pair in $\mathscr{H}_1^2$ represents LR.

Remark 7.7 (Some regularity is indispensable). (P), (N), (CM) and (LI) alone do not force continuity. Using a Hamel basis one obtains $\rho = \mathbb{E} + \varphi \circ I_k$ with $k(t) = t(1-t)$ (so that I_k vanishes on constants and is cash invariant) and φ additive but discontinuous. Then $[\rho : \mathbb{E}]$ satisfies (P), (N), (CM) and (LI), hence ($\mathsf{P}_\exists$) and (R), and is not of the form in Theorem 6.1. This witness uses the axiom of choice; all the others are constructive.

Example 7.8 (Independence of (RM) in Theorem 6.2)**.** Let $h(t) = t + \frac{1}{4}\sin 2\pi t \in \mathscr{H}_1$ (Figure 3), so $h(0.4) \approx 0.547 > h(0.6) \approx 0.453$, and $\mathrm{LR} = [I_h : \mathbb{E}]$. All other axioms hold ((RC) by the Lipschitz bound), but the implied risk measure I_h is not monotone, and no distortion pair represents LR because the pair is unique.

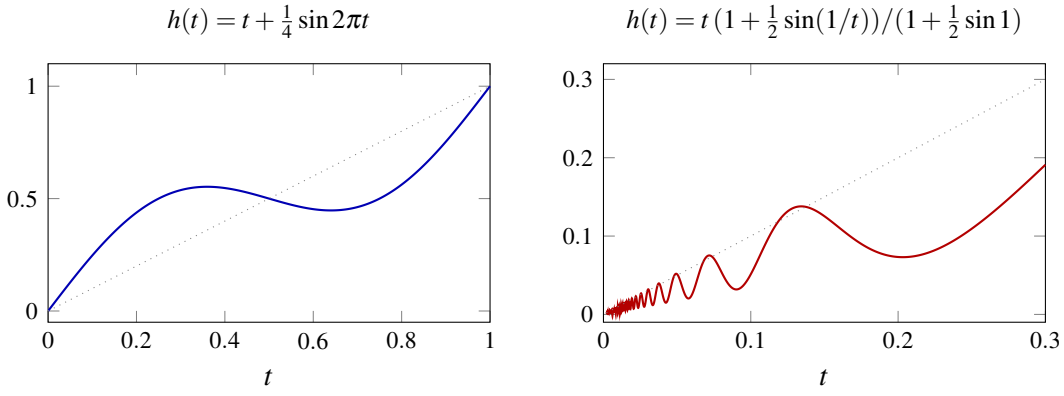


Figure 3: Numerator distortions of the witnesses for (RM) (left: of bounded variation, not monotone) and for (RC) (right, near 0: continuous, not of bounded variation). The dotted line is the identity.

Axioms (A0) and (A1) define ρ and Π on $\mathscr{X} \setminus N_0$, through which the remaining axioms are stated, so they are not treated as independent axioms. Two examples illustrate the identification they provide. If $\mathrm{LR}_0 \equiv 1$, (A0) fails and $(\mathrm{id}, \mathrm{id})$ and $(\sqrt{\cdot}, \sqrt{\cdot})$ both represent LR_0. If $\mathrm{LR}_0 = (C_h/C_g)^3$ with distortions $h \neq g$ and $g > 0$, then N_0 is that of C_h/C_g, but (A1) fails, because the cash line has a pole of order three at $-C_g(X)$ for $X \notin N_0$.

Remark 7.9 (Atomlessness). Atomlessness is used to attain every $p \in [0,1]$ as $P(A)$. On a space with atoms, (h, g) is identified only on the range of P.

8 The case study: report of the AI process

This section reports how the result of Sections 3–7 was obtained with AI, following the items of the standard in Table 2; the case study is in tier (iii). The tools are identified in the declaration before the references (item 1). The section describes the organization of the work (items 2 and 3), the knowledge base (item 4), the errors found (item 5), the guardrails (item 6), the formal check in Lean (item 7), the revisions prompted by readers, and the verification status (item 8). The records (item 9) are in the repository named in the code availability statement.

8.1 Organization of the work

The work used Claude (Anthropic) through the Claude desktop application in Cowork mode, which runs the assistant in a sandboxed environment and can coordinate several AI agents in a workflow; the later revisions, including the second panel of Section 8.7, used Claude Code in the author's code editor, which ran the panel as a scripted workflow. AI agents worked on designated parts of the project, under the author's direction. A main assistant organized the work and drafted and revised the manuscript. A roundtable of six agents, five of them each assigned a strand of the literature and one acting as chair, surveyed the axiomatic theory of premiums, risk measures and performance ratios and produced a synthesis. A larger panel of agents covered ten literature strands, proposed candidate statements, ranked them by means of judge agents, drafted proofs, and subjected the proofs to separate AI-assisted checking runs over several rounds. The literature review was conducted by these AI assistants under the author's supervision: the author defined the strands to be covered and the questions to be answered, and selected and framed the material presented in Section 3. Some agents were given the names of researchers as personas, to represent the perspective of a literature strand; these were simulations, and the researchers concerned did not participate, were not consulted and bear no responsibility for the content. The research questions, the choice of formulation and all decisions on the content of the paper were made by the author.

Figures 4 and 5 show the organization of the work. In Figure 4 the author assigns the task of each AI component and reviews its output. Figure 5 shows the internal pipeline of the panel, in which candidate statements and proofs pass through ranking and repeated critical review before being returned to the author. Figure 6 gives a schematic picture of the panel: the ten strand agents around a shared digest, the proof bench of judge, prover and checking agents, and the author's place in the loop.

AI assisted with the following tasks:

- literature exploration;
- formulation of mathematical statements and drafting of proofs;
- construction of counterexamples and examples;
- manuscript drafting and revision.

8.2 How the knowledge base for this paper was assembled

The agents draw on two kinds of knowledge: what their underlying models learned in training, which can be recalled only approximately, and documents read into their working context during the task. In this project the second kind was organized as follows.

The five strands of the roundtable were identified by the author: premium principles and capital allocation; conic finance and performance ratios; coherent risk measures and acceptance sets; axiomatic

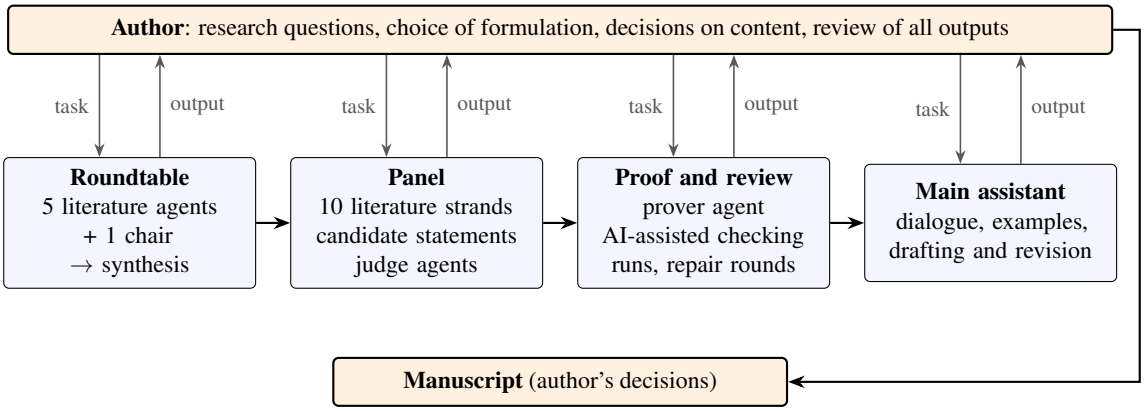


Figure 4: Organization of the work. The author assigns tasks to each AI component and reviews every output; the manuscript reflects the author's decisions. The stages were used in the order shown, with returns to earlier stages when the author redirected the work.

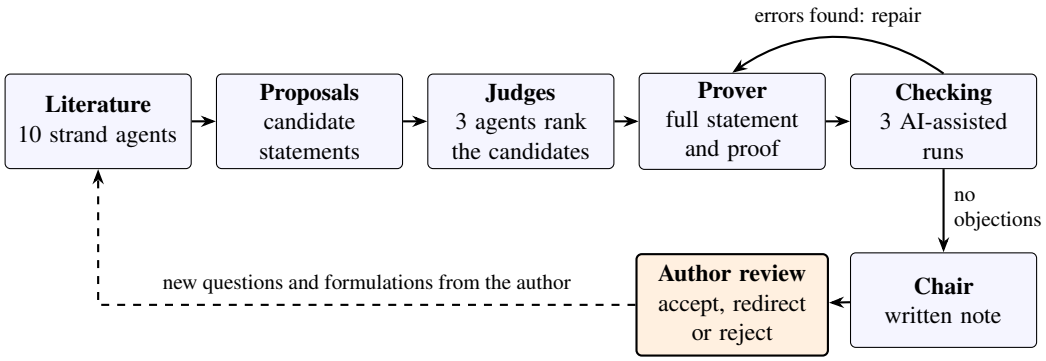


Figure 5: Pipeline of the panel. Literature agents survey their strands; proposal agents formulate candidate statements, which judge agents rank. A prover agent drafts a full statement and proof, which separate AI-assisted checking runs examine; reported errors are sent back for repair over several rounds. A chair agent writes up the result for the author, who accepts it, redirects the work or rejects it. Agreement among checking runs does not establish correctness.

risk measures and elicitability; and the extended-loss-ratio framework with its association. Its synthesis served as the starting point of the panel.

The panel divided the literature into ten strands: insurance-pricing axiomatics; Choquet integrals and comonotonic additivity; signed Choquet integrals; law-invariant coherent and convex risk measures; acceptability indices and quasi-convex duality; return and star-shaped risk measures; elicitability; performance-ratio axiomatics; decision theory and functional equations; and a critique of the earlier synthesis. Each strand agent returned exact statements of if-and-only-if theorems with their hypotheses, a confidence label (high, medium or low) and a note on its source. For example, the signed representation results of Cerreia-Vioglio et al. (2012, 2015) were taken from their working paper versions.

There was no separate retrieval index. Instead, the strand reports were merged into a written digest of about 415,000 characters, which, together with the ten candidate formulations (about 245,000 characters), formed the shared knowledge base of the later stages: the judge and prover agents were instructed to read these files in full before working, and Section 3 was written from the digest. The confidence labels therefore travelled with each statement into the stages that used it.

In addition, the author used Gemini (Google) to prepare summaries of the key works cited in Section 3: Gerber (1974), Yaari (1987) and Wang et al. (1997) on premium principles; Murofushi et al. (1994), Cerreia-Vioglio et al. (2012, 2015) and Wang et al. (2020a) on signed Choquet integrals; Artzner

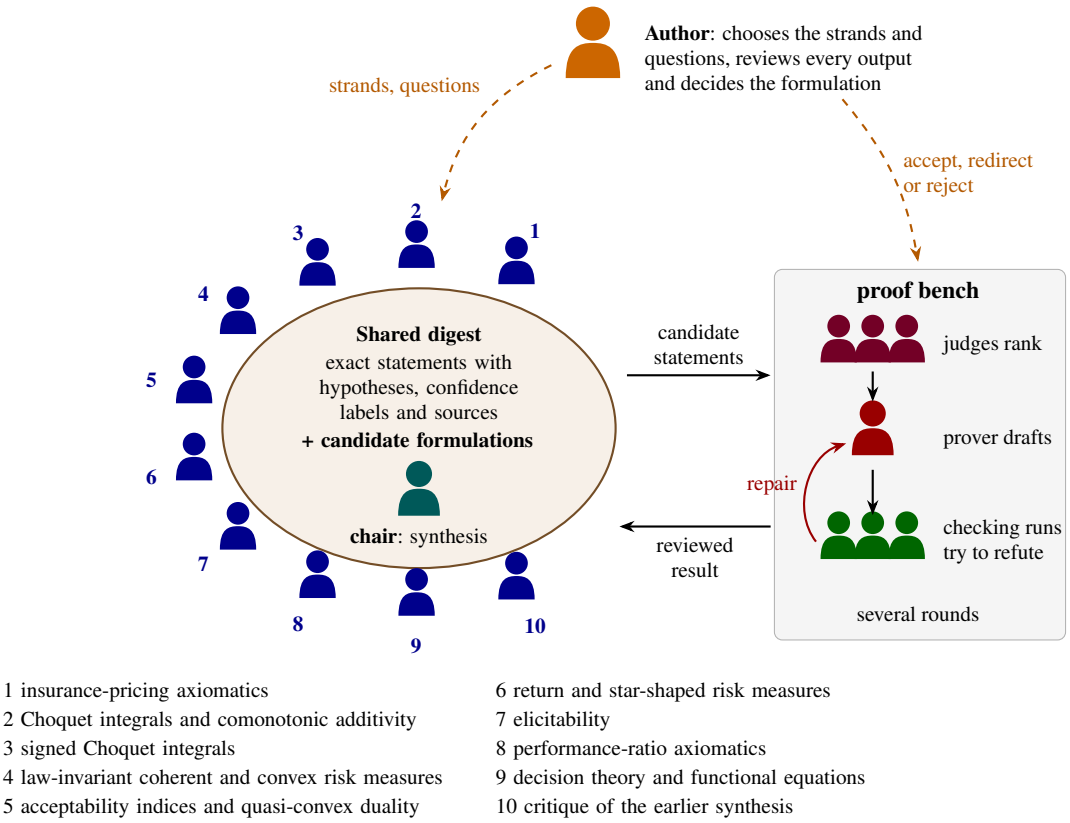


Figure 6: The panel at work. Ten literature agents (numbered as in the key), each assigned a strand by the author, write exact statements into a shared digest, which together with the candidate formulations forms the knowledge base read in full by the later agents; a chair agent synthesizes. Candidate statements pass to the proof bench, where judge agents rank them, a prover agent drafts proofs and separate checking runs try to refute them, with errors returned for repair. The author sets the strands and questions and accepts, redirects or rejects each result. The icons are schematic: the agents are separate runs of a language model.

et al. (1999) and Kusuoka (2001) on coherent risk measures; and Cherny and Madan (2009), Chew (1983) and Dekel (1986) on performance ratios and ratio representations. Each summary stated the core results of the work and their role in this paper. The descriptions in Section 3 and the uses of these works in Sections 5, 6, 7 and S5 were compared with these summaries; no discrepancy was found, and three descriptions in Section 3 were made more precise. Since the summaries are themselves AI output, this comparison is a consistency check, not a check against the original publications.

Later, the author supplied author versions of three cited papers: Wang et al. (2020a) (working paper of June 2019), Wang et al. (2020b) (version of May 2020, with a correction of May 2023) and Han et al. (2025) (version of June 2024). The statements taken from them were checked against these versions: the definition of I_h, the characterization theorem used in the representation step and the convexity theorem of Wang et al. (2020a); the monotonicity criterion and Lipschitz bound, which are also stated there; the extension to general spaces, the convention $\mathrm{ES}_1 = \operatorname{ess\,sup}$ and the mixture representation of Wang et al. (2020b); and the axioms and the two characterization theorems of Han et al. (2025). No discrepancy was found. Lemma numbers that could differ between these versions and the published articles were removed, and the description of Han et al. (2025) in Section 3 was rewritten from the paper.

The account of OpenAI's work and of the responses of the mathematical community in Section 2.1 was assembled later, from two independent AI-prepared reports (see the Discussion in Section 2.2). Statements were kept where the two reports agree or where a primary source was read directly (the

GitHub repositories, the two arXiv papers and the AGMAI guidelines); OpenAI's own pages could not be retrieved automatically; the texts of its Navier–Stokes and First Proof posts, and a localized version of its October post on academic collaborations, were retrieved by the author in a browser, and the details of those projects are taken from them; the October post is paraphrased rather than quoted.

8.3 Hallucinations and semantic errors encountered

Statements about the literature. The main risk of hallucination concerned the exact hypotheses, conclusions and numbering of cited results. Two entries of the digest disagreed on whether the mixing measure in the Kusuoka representation includes an atom at the endpoint; the paper therefore states the mixture over $[0,1]$ with an explicit convention (Corollary S5.4(a)) and cites the representation without theorem numbers. For the same reason, the representation step used in the proofs is not taken from the literature but reproved (Lemmas S1.7–S1.10).

Errors in mathematical content. Table 5 lists the errors in AI-generated mathematics that were found during the project. Most are semantic: the statement proved, or claimed, differed from the statement intended, typically at a degenerate case (zero premium, the break-even set, the labels ∞ and $\star$) or in the scope of a claim.

8.4 Guardrails applied in this project

- *Labels on knowledge.* Every literature statement in the digest carried a confidence label and a note on its source, as described above.
- *Independent summaries.* Summaries of the key cited works prepared with Gemini were compared with their descriptions and uses in the paper.
- *Self-contained proofs.* Results whose exact statements were uncertain are reproved; in particular the representation of comonotonic additive functionals, for which the proof cites Wang et al. (2020a, Thm. 1), is reproved in Lemmas S1.7–S1.10.
- *Adversarial checking.* The three checking runs on the panel's proofs were instructed to refute; one tested the degenerate case $\mathrm{LR} \equiv 1$ with a premium that is not comonotonic additive, and two recomputed the numbers in the counterexamples. The formulation through the implied premium and risk measure (Section 5 and Section S1 of the Supplement) was given to a further checking run with instructions to look for circularity and for the cases $\Pi(X) = 0$, break-even books, $\mathrm{LR} \equiv 1$ and $c = 0$.
- *Repair loop.* Reported gaps were returned to the prover agent; the panel's proofs went through two repair rounds before the final checking round.
- *Counterexamples.* Each axiom is tested against an explicit loss ratio (Section 7).
- *Recomputation.* The values in Examples 6.5, 6.6, S5.5 and in the counterexamples were recomputed by script, and the monotonicity range in Example S5.2 was confirmed on a grid.
- *Formal checking.* The loss-ratio-specific steps were formalized in Lean and compiled; see Section 8.6.
- *A second system.* The review with ChatGPT, a model from a different provider, found three of the errors in Table 5 that the earlier checks had missed.

8.5 What the prover produced and what was formally checked

The "prover" of Figure 5 is a language-model agent that writes proofs in ordinary mathematical language; it is not a formal theorem prover. In this project it drafted the first complete if-and-only-if theorem, in a formulation based on an assumed cash-line axiom, with proofs of the results that are now Theorems 6.1, 6.2, S3.1, S4.3 and S4.4 (the last three in the Supplement) and of the supporting lemmas; the final checking round of these proofs returned two reports without objections and one report of gaps

Error	Found by	Resolution
$\mathbb{E}/C_g$ with concave g claimed quasi-convex (early synthesis)	panel agents	quasi-concave; Cor. S5.4(d)
quasi-convexity stated as an if-and-only-if in four candidate formulations	judge agent ($h=t$, $g=\sqrt{t}$, $\beta=2$)	only sufficient conditions kept
candidate route via Sincov and Cauchy equations	judge agent (continuous h not of bounded variation)	route abandoned
main argument claimed to apply verbatim when (ρ,Π) is only assumed to exist	checking run	separate argument given
association claimed to hold exactly on the finite domain	checking run	holds off $\{\rho=\Pi=0\}$; Cor. S5.1(a)
"$h-g$ concave" as a sufficient condition for quasi-convexity	main assistant, during rewriting	h concave, g convex
converse of Theorem 6.1 failed for (LI) (labels unconstrained)	checking run	(LI) compares finite values only
claimed equivalence with the cash-line formulation	checking run	convention (S) added; Remark S2.2
regularity axioms undefined for two counterexamples	checking run	(RC), (RM) stated for the price point and the zero of the cash lines
uniqueness of the extension in Prop. S4.5	review with ChatGPT	unique up to labels, unique under (S)
independence claimed for (A0) and (A1)	review with ChatGPT	treated as illustrations; Sec. 7
mixture of expected shortfalls missing the endpoint	review with ChatGPT	$\mathrm{ES}_1 :=$ ess sup; Cor. S5.4(a)
Revision of Section 8.7		
implied risk measure identified before (CM) was derived (circular citation)	adversarial agent of the panel	identified after Step 3; Lemma S1.1
regularity axioms in a multiplied form excluded the witness of Example 7.3	two adversarial agents	(RC), (RM) use only the predicates "not finite", "$=0$", "$=1$"
test of "$\mathrm{LR}(W+c)=1$ for all $c\neq 0$" by sampling c gave false results on break-even books	adversarial agent	finite predicate $\mathrm{LR}(W+1)=1$
failure of (R) for $[\Pi':\Pi']$ claimed from random trials, which never hit the pole	adversarial agent	explicit books; Example 7.4
non-finite point of $[I_h:I_g]$ misplaced in the example for ($\mathsf{P}_\exists$)	main assistant, on rereading	non-finite at $X=0$

Table 5: Errors in AI-generated mathematics found during the project, and their resolution.

in side claims, which were repaired. The present formulation through the implied premium and risk measure (Lemmas S1.3–S1.6 and Proposition 5.13) was drafted afterwards by the main assistant and checked by one further run, whose findings appear in Table 5.

The loss-ratio-specific steps were also checked formally in Lean (Section 8.6).

8.6 Formal checking in Lean

Lean and what a formal check establishes

Lean 4 (de Moura and Ullrich, 2021) is an interactive proof assistant: a programming language in which definitions, statements and proofs are written in a formal language based on dependent type theory, and in which every proof is checked by a small trusted program, the kernel. A proof may be written step by step or with the help of tactics, procedures that search for parts of a proof automatically, but whatever a tactic produces is checked again by the kernel. Mathlib (The mathlib Community, 2020)

is the community library of formalized mathematics for Lean; it supplies the real numbers, groups, limits and the other notions used below.

Two features of Lean are used to state precisely what has been checked. A proof can be left unfinished by writing `sorry`, which Lean accepts with a warning; a finished development contains none. The command `#print axioms` lists the foundational axioms on which a theorem depends; ordinary classical mathematics in Lean uses exactly three of them, `propext` (propositional extensionality), `Classical.choice` (the axiom of choice) and `Quot.sound` (soundness of quotients), and an unfinished proof shows up in this list as `sorryAx`.

A successful check establishes that the formal statements follow from the formal definitions. It does not establish that the formal definitions and statements say what the paper says, which must be checked by a human reader, and it says nothing about results that enter as hypotheses or are not formalized.

For this paper a formal check is useful for three reasons. First, the proofs were drafted with AI assistance, and the errors recorded in Table 5 show that such drafts can contain steps that read well and are wrong; the kernel does not depend on how a proof was produced. Second, the steps that are specific to loss ratios (Lemmas S1.3–S1.6) are elementary but delicate, with case distinctions at $\Pi(X) = 0$, at break-even books and at the price point, which is where informal checking is most likely to miss a case. Third, formalization forces the hypotheses to be stated exactly, and here it shows that those steps use only a few algebraic properties of comonotonicity and equality in law (see "What was formalized" below).

What was formalized

The formalization works in an abstract setting. Positions form an additive commutative group $\mathscr{X}$ with an additive embedding $c \mapsto \mathrm{const}(c)$ of the sure losses, a relation "comonotone" under which every position is comonotone with every constant and which is preserved when both positions are shifted by constants, and a relation "equal in law" preserved by a common cash shift. A loss ratio is a map from $\mathscr{X}$ to $\mathbb{R} \cup \{\texttt{none}\}$, where `none` stands for a non-finite value, so that the labels ∞ and $\star$ are identified as in (LI). The axioms read as follows; `prem` is the implied premium and `rhoD` is $\Pi \cdot \mathrm{LR}$.

```
def AxP : Prop := ∀ x : X, ∃! c : ℝ, LR (x + S.const c) = none
def AxN : Prop := ∀ c : ℝ, c ≠ 0 → LR (S.const c) = some 1
def AxLI : Prop := ∀ x y : X, S.eqd x y → LR x = LR y
def AxCM (hP : AxP S LR) : Prop :=
  ∀ x y : X, S.comon x y → LR x ≠ none → LR y ≠ none →
    prem hP x + prem hP y ≠ 0 →
    LR (x + y) = some ((rhoD hP x + rhoD hP y) / (prem hP x + prem hP y))
```

These coincide with Axioms 5.6, 5.9, 5.10 and 5.18, with one difference of form: `AxCM` is stated with the products $\Pi \cdot \mathrm{LR}$ over the sum of the premiums, which is Axiom 5.10 written as in its second expression, so that the relative-cost weights $w_{X,Y}$ do not appear in the Lean text. The file has about 500 lines with 10 definitions and 26 lemmas and theorems. The axioms ($\mathsf{P}_\exists$) and (R) of Section 5, their equivalence with (P), (N) and (CM) (Lemma S1.1), and the regularity axioms in the form of Lemma S1.2 are not yet formalized. Table 6 lists the results of the paper that are formalized.

The theorem `main_conditional` takes the representation step as a hypothesis: any cash-additive, comonotonic additive, law-invariant and regular functional is of the required form (Wang et al., 2020a, Thm. 1). From (P), (N), (CM) and (LI) it derives that the implied premium and the implied risk measure satisfy the hypotheses of that step and that $\mathrm{LR} = [\rho : \Pi]$ everywhere. Not formalized are the representation step itself (Lemmas S1.7–S1.10), which would require a theory of Choquet integrals that Mathlib does not provide, the fact that L^∞ with comonotonicity and equality in law is an instance of the abstract setting, the layer formula, and the examples and counterexamples.

Result in the paper	Lean name
Implied premium, uniqueness and cash additivity (Lemma S1.3)	`prem`, `prem_shift`, `prem_const`
Implied risk measure and cash additivity (Lemma S1.4)	`rho`, `rho_cash`
Möbius form and residue (Proposition 5.13(a)–(c))	`mobius`, `rho_eq`, `residue`
Comonotonic additivity, also on the break-even set (Lemma S1.5)	`comon_additive`
Law invariance of the implied pair (Lemma S1.6)	`LI_transfer`
Converse construction and uniqueness of the pair	`converse`, `unique_pair`
Association (Corollary S5.1(a))	`association`
Pooling at non-finite values (Corollary S5.3)	`pooling_extended`
Algebraic core of Proposition 5.21	`no_monotone_core`
Theorem 6.1, given the representation step	`main_conditional`

Table 6: Results formalized in Lean.

How the check was carried out

The Lean file was written by the main assistant and compiled on the author's computer, since Lean could not be installed in the assistant's cloud workspace. The assistant read each build log, corrected the file, and the author ran the next build; the third build succeeded. The build-and-repair loop and a timed record of every step are given in Section S7 of the Supplement.

Checks performed

The following checks were made to ensure that the outcome is what is reported.

- *No change to statements during repair.* The corrections after builds 1 and 2 changed proof steps only; no definition and no theorem statement was altered.
- *No unfinished proofs.* The final file contains no `sorry`, no `admit` and no added `axiom`.
- *Axiom report.* `#print axioms` for each of the ten theorems in Table 6 lists only `propext`, `Classical.choice` and `Quot.sound`.
- *Clean rebuild.* The final build started from an empty build directory, so its result does not rely on cached output.
- *Identity of the file.* The file compiled in the final build was compared with the file reported here and found identical.
- *Fidelity of statements.* The formal axioms and theorem statements were compared with the paper by the assistant; this comparison is to be repeated by the author before the code is released.

The remaining warnings concern style and the use of renamed library lemmas; they do not affect correctness. The Lean source and the build log are available in the repository `https://github.com/hirbod-assa/loss-ratio-lean`, which contains the Lean project (Lean and Mathlib version 4.36.0-rc1) together with instructions for rebuilding it with `lake build`.

8.7 Revisions prompted by readers

Three comments from readers of earlier versions changed the paper. They are reported here because they show where the direction of the work came from, and because each led to further work with AI agents that the standard asks to be recorded.

A public comment on the research question. A reader of the announcement of an earlier version questioned whether an accounting ratio deserves an axiomatic characterization. The exchange that followed between the author and the main assistant is summarized in the Discussion in Section 1; it led

to the motivation from risk reporting given there and changed no mathematics.

Axioms on the loss ratio alone. In an earlier version the pooling axiom (CM) and the regularity axioms were stated through the implied premium and the implied risk measure. Peng Liu, who read that version, found the characterization clean but remarked that some axioms were therefore based on derived quantities, and that it would be preferable to state all axioms explicitly on the loss ratio itself. The present formulation, in which every axiom refers only to values of LR (Section 5.1), was obtained in three steps.

Exploration by the author with the main assistant. The author first tried to replace the implicit premium by explicit formulas in terms of LR; these attempts, and the reason they cannot suffice, are recorded in the Discussions after Proposition 5.15. An intermediate version stated (CM) with weights given by the points at which the cash lines are not finite. The author rejected it: such a point is the premium under another name, and the requirement was that no axiom use the premium, a risk measure or a point read off a cash line.

The panel. The author then asked for a panel to replace (CM). It was run as a scripted workflow of nine agents. Four proposal agents worked from different angles: stability of level sets under comonotone pooling together with a condition on the shape of the cash lines; a single pooling identity on cash lines; elimination of the weight from (CM); and the limits of what a pure loss-ratio axiom can achieve. Each proposal was passed to its own adversarial agent, instructed to refute it and to recompute its numbers. All four survived with repairs; the errors the adversarial agents found are in the last part of Table 5. A judge agent ranked the four, chose the proposal that became ($\mathsf{P}_{\exists}$) and (R), adopted the form of the regularity axioms from another proposal, and collected the impossibility results that became Proposition 5.3 and the witnesses of Section 7. The judge also checked the combined system in exact rational arithmetic on a four-point space for three representing pairs, a pair of distortions, $h = g$ and a signed pair: (R) held on every test pair (600 for each of the first and third, 450 for $h = g$), including, for the first, 75 pairs of non-constant break-even books, 75 mixed pairs and 150 pairs whose premiums cancel; the sure amounts in (RC) and (RM) equalled I_g and I_h; and each witness of Section 7 failed. The prompts of the workflow, the judge's report and its script are kept with the records of the project (`ai_process/`). The main assistant then wrote Section 5.1, Lemmas S1.1 and S1.2 and the changes in the other sections, and repeated the numerical checks.

The author's decisions. The author adopted the judge's recommendation and questioned the remaining hypothesis of (R) (the last Discussion in Section 5.1); the answer became Remark 5.4 and Proposition 5.3. The nonnegative case (Section S4 of the Supplement) was left in its earlier form. The proofs of Lemmas S1.1 and S1.2 and of Proposition 5.3 were checked by one adversarial agent, by the judge and by computation; they are not formalized in Lean and were not part of the review with ChatGPT.

The structure of the paper. An editor of an actuarial journal, to whom the author sent an earlier version for an opinion on the reporting standard, suggested that the paper was long, that it was structured around the mathematical contribution with the report on AI as a detailed acknowledgement, and that the standard should instead be a contribution in its own right, with the mathematical result as its case study; the editor also noted that faster derivation tends to produce more results and examples than a reader needs. The comments on the standard itself are summarized in the last Discussion of Section 2.6. In response, the paper was restructured with the main assistant: the standard and the comparison with large-scale efforts now come first (Section 2); the literature review was shortened; the proofs, the version without law invariance, the version for nonnegative losses, the consequences of the representation and the record of the Lean check moved to the Supplement; and the Discussions recording the author's exchanges with the main assistant were added. The mathematical statements were not changed in this step.

8.8 Verification status

The proofs have undergone AI-assisted critical review. Such review is fallible and does not constitute formal verification; this applies equally to the informal review with ChatGPT. No claim of machine-checked correctness is made beyond the results formalized in Lean, in the abstract setting and with the hypothesis described in Section 8.6; those results concern the formulation with (P), (N) and (CM), and the equivalence with the present axioms (Lemma S1.1) is not formalized. Independent human review of all proofs has not yet taken place; the comments reported in Section 8.7 concerned the formulation of the axioms and the structure of the paper. Apart from the three papers named in Section 8.2, the references have not been checked against the original publications. Section 2 and the comparison with OpenAI's work rely on the cited public sources; where these are press reports, this is stated.

8.9 What the case study shows about the standard

Four observations from the case study motivate the items of Table 2. First, the errors were found by different checks: checking runs within the panel, adversarial agents, the main assistant during rewriting, and a review with a model from a different provider, which found three errors that the earlier checks had missed (Table 5). Reporting the checks, and not only their conclusion, tells the reader which kinds of error have been looked for. Second, the decisive steps came from the author: the reformulation through the implied premium and risk measure came from an objection to an AI-drafted formulation that was internally correct but assumed what should have been proved, and the axioms on the loss ratio alone came from insisting, after a reader's remark, that no derived quantity appear in an axiom. A record of the division of labour makes such contributions visible. Third, the revisions after the Lean check changed the axioms, so that the formal check now covers an equivalent earlier formulation rather than the present one; a verification status stated part by part, with dates, makes this visible instead of hiding it. Fourth, agreement among AI reviews does not establish correctness, and a formal check covers only what was formalized.

9 Discussion and open problems

The standard. The proposed standard asks authors of AI-heavy work for information that a reviewer can use, in proportion to how heavily AI was used, and it separates what can be verified from what must be declared. Several questions are open. Where exactly the boundaries between the tiers lie will become clear only from use. How reviewers may themselves use AI, and whether the verification of formal statements against the paper can be shared between author, reviewer and the community, are questions for journals. The standard also presupposes that formal verification is feasible for part of the work; for risk management this depends on libraries of formalized probability and Choquet integration that do not yet exist. Finally, the comparison with large-scale efforts suggests that the design principles of multi-agent work, and the records needed to report it, are largely independent of scale, but this is based on one case study.

The case study. A loss ratio carries its own premium and its own risk measure. The premium is the price point of the cash line, the sure payment at which the book changes hands, and the risk measure is loss ratio times premium, visible even where the ratio explodes as the residue of the singularity. With this reading, axioms on the values of the loss ratio alone, whose substance is that the underwriting results of comonotone books of zero premium add, together with law invariance and regularity of two sure amounts, characterize the loss ratios that are ratios of two Choquet-type functionals. The characterization is sharp, and the pair is unique. At the level of representations it is a transfer of the theory of Yaari (1987), Schmeidler (1986), Wang et al. (1997) and Wang et al. (2020a); at the level of the loss ratio it is new, and it comes with facts that have no single-functional counterpart: no non-trivial loss

ratio is monotone (Proposition 5.21), the association with a family of risk measures holds exactly off $\{\rho = \Pi = 0\}$ and is signed (Corollary S5.1 in the Supplement), and no axiom on finite values of the ratio can replace the pooling axiom (Proposition 5.3).

Open problems.

(1) *Beyond comonotone pooling.* The most promising direction is to drop (R) in favour of an axiom that allows diversification, so that the implied premium and risk measure are convex or subadditive rather than comonotonic additive. The implied premium and the implied risk measure are defined without (R) only partly (the premium needs only (P); the accounting reading of ρ needs pooling with sure losses), and the resulting class, a ratio of Kusuoka-type functionals, has no characterization yet. Replacing comonotonic additivity by convexity leads to suprema of signed Choquet integrals (Liu et al., 2020), so no single pair (h, g) and no uniqueness should be expected.

(2) *Weaker cardinal axioms.* Can (R) be replaced by the mediant inequality plus a supplementary condition? By Section 7, such a condition must involve non-finite values of LR. The inequality implies additivity for comonotone pairs strictly on the same side of break-even, but not for pairs straddling 1 or lying in the break-even set.

(3) *Order properties of* LR. (RM) is an order property of LR, but of the sure amounts at which its cash lines become non-finite or vanish. Whether it is equivalent to an order property of the values of LR alone is open; by Proposition 5.21 it cannot be pointwise monotonicity.

(4) *Loadings on sure losses.* We conjecture that an analogue of Theorem 6.1 holds for ratios $[I_h : \kappa I_g]$ with a known loading $\kappa > 0$, which would admit constant loss ratios $\neq 1$.

(5) *Quasi-convexity.* Necessity of the conditions in Corollary S5.4(d) is open.

(6) *Actuarial domain.* The axioms for nonnegative losses (Section S4 of the Supplement) are not yet stated on values of the loss ratio alone. Whether (A4) imposed only on pairs with $X, Y, X+Y \notin N_0$ suffices is open; a truncation-continuity (Greco-type) version of the signed tier on the cone is not known.

(7) *Unbounded losses.* Extensions to L^p and Orlicz spaces via Wang et al. (2020b) are not treated.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During this research and the preparation of this manuscript, the author used Claude (Anthropic), accessed through the Claude desktop application in Cowork mode between 20 and 25 October 2026, and through Claude Code in a code editor between 26 October and 9 November 2026, with the model identifiers `claude-sonnet-4-6`, `claude-fable-5-1`, `claude-sonnet-5-5` and `claude-opus-5-5` recorded in the available session logs, to assist with literature exploration, formulation of mathematical statements, construction and criticism of proofs and counterexamples, writing a Lean formalization of part of the proofs, comparing the approach with published multi-agent work, and manuscript drafting and revision. The author also used ChatGPT (OpenAI) for an informal review of the manuscript's proofs, which identified potential errors and qualifications and suggested revisions to the mathematical statements and to this disclosure; this review did not constitute formal proof verification. The author used Gemini (Google) to prepare summaries of the content of key cited works, against which the descriptions of the literature in the manuscript were compared. Substantive mathematical assistance and checking procedures are described in Section 8. AI-generated arguments and references were treated as proposals requiring verification. The author takes full responsibility for the final mathematical statements, proofs, references and presentation.

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Code and data availability

The Lean 4 formalization described in Section 8.6, the source of the paper and of the Supplement, the records of the AI process, and the skill and agent files of Section 2.7 are available at https://github.com/hirbod-assa/loss-ratio-lean.

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